



MRC
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Simulation studies as a tool to assess and compare the properties of statistical methods – an overview

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Hello



We work at the MRC Clinical Trials Unit at UCL, an academic trials unit with a around 60 statisticians.



Our work focuses on the development, evaluation and understanding of **statistical methods** for health research

About today

- We assume you have some familiarity with simulation studies
- We will nonetheless cover some more introductory/basic points
- Today, we are focused on interesting and tricky concepts
- Basics are covered in the Stats in Med tutorial

Acknowledgements

Thanks to **Anne-Laure Boulesteix** and the **IBS-DR** for the invitation, to **Willi Sauerbrei** for suggesting this session and chairing (in-person), and **Georg Heinze** for chairing (remote)

Acknowledgements

**Ian
White**



Tra My Pham



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Quartagno**

Part 1: planning (Tim)

- **Introduction and overview**
- Neutral comparisons?
- Phases of methodological research
- Elements of a simulation study: ADEMP

Introducing simulation studies

Simulation studies are used in a variety of disciplines.

They are an important and useful tool and enable researchers to understand the properties of methods and to explore how methods compare.

What do we mean by ‘simulation studies’?

By ‘simulation studies’, we mean studies of the property of some statistical method.

In a typical simulation study, we might:

1. Generate data from a known distribution (so that we know the ‘truth’)
2. Analyse the data
3. Compare the analysis results with the truth

What do we not mean?

Simulation is often used for many specific analyses: MCMC methods, multiple imputation, bootstrapping. etc.

These uses of statistical simulation are **not** the topic of this workshop.

The breadth of simulation studies

Simulation studies can be used to:

- Check that code does the intended analysis
- Check robustness of programs
- Better understand statistical concepts
- Understand new commands
- Check/reassure oneself of mathematical results
- **Evaluate new or existing statistical methods**
- **Compare methods head-to-head**
- Calculate sample size / power

and more...

To consider...

If your mind wanders during this session, think about whether a simulation study can meaningfully address the following questions:

1. Is the mean of a random sample unbiased for the population mean?
2. How does linear regression perform when the residuals are not independent, not normal or not homoscedastic?
3. Should missing data be analysed assuming MAR or MNAR?

Part 1: planning (Brennan)

- Introduction and overview
- **Neutral comparisons?**
- Phases of methodological research
- Elements of a simulation study: ADEMP

Neutral comparisons in simulation studies: what do we mean by 'fair'?

Idea of 'fair' comparison is intrinsic to scientific research

We don't want to compare healthy patients who get treatment vs. unhealthy patients who get control

Same is true in methods research

Want comparisons to be 'fair'

Concept of 'fair' is simple in healthcare studies, e.g. the causal difference between treatments (though can be **hard** to estimate!)

But the concept is trickier for methods research



RESEARCH ARTICLE |  Full Access

A fair comparison of tree-based and parametric methods in multiple imputation by chained equations

Emily Slade , Melissa G. Naylor

First published: 29 January 2020 | <https://doi.org/10.1002/sim.8468> | Citations: 14

Motivation

Literature has suggested that **nonparametric tree-based imputation methods outperform parametric methods** in terms of bias and coverage when there are interactions or other nonlinear effects among the variables. However, **these studies fail to provide a fair comparison as they do not follow the well-established recommendation that any effects in the final analysis model (including interactions) should be included in the parametric imputation model.**

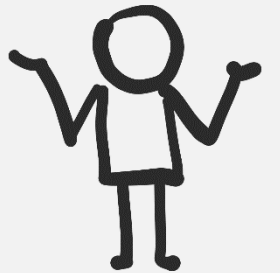
Findings after a “fair” comparison

We show via simulation that properly incorporating interactions in the parametric imputation model leads to much better performance. In fact, correctly specified parametric imputation and tree-based random forest imputation perform similarly when estimating the interaction effect.

Parametric imputation leads to slightly higher coverage for the interaction effect, but it has wider confidence intervals than random forest imputation and requires correct specification of the imputation model

Fair = best version of each method against each other?

I don't know much about multiple imputation – I probably wouldn't have the knowledge to “tune” the parametric models in the way described by Slade and Naylor

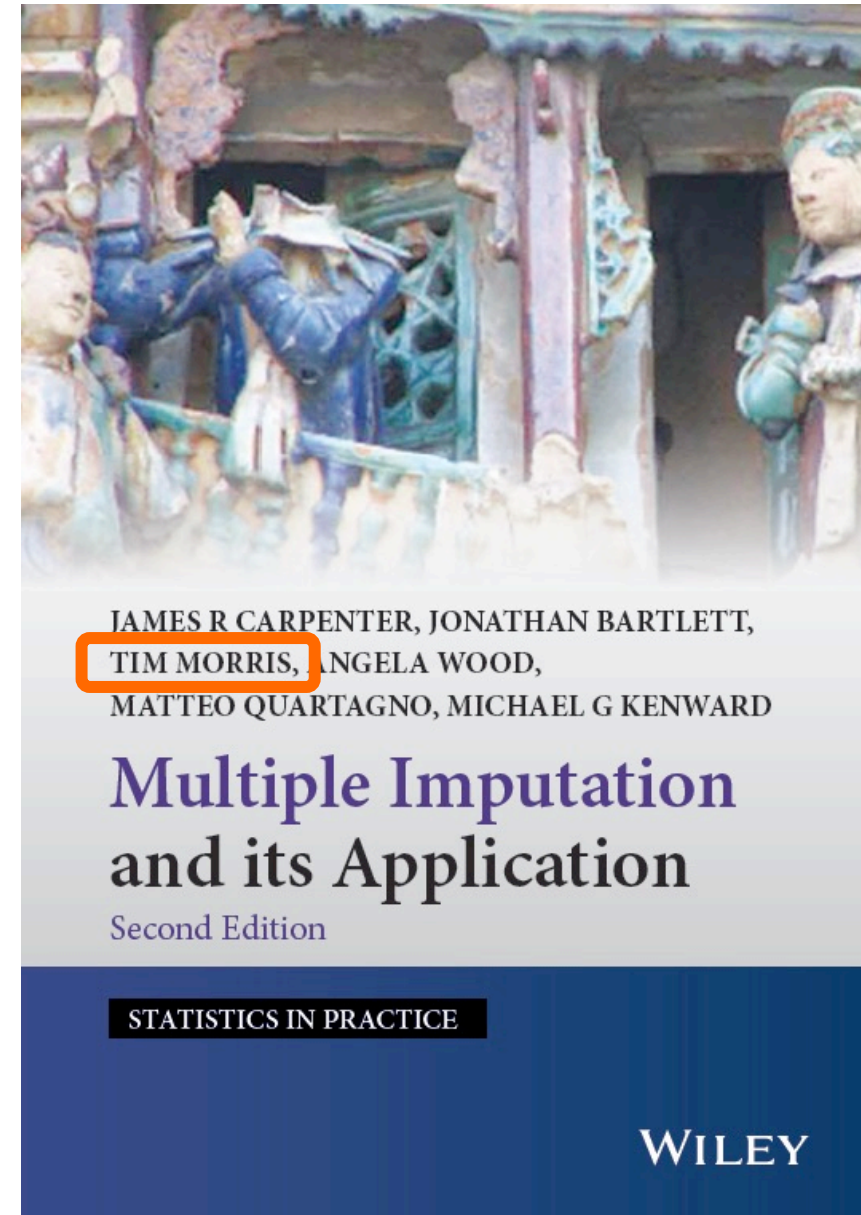


Fair = best version of each method against each other?

Tim knows a bit about multiple imputation

(available from all good book stores near you)

doi.org/10.1002/9781119942283



Who is the comparison for?

- Links to the concept of *estimands* (i.e. research objectives)
- Concept of estimands has been used for decades, but gaining recent attention in RCTs/observational studies to clarify target treatment effect
- Key point: it's not enough to say “we’re comparing treatment A vs. B”
- Need to specify *how* treatments are being used
 - Treatment A vs. B, regardless of non-adherence?
 - Or treatment A vs. B, if everyone took treatment as prescribed?
- These answer *different* questions, and require *different* methods

What's the research question?

The same issue applies in simulation studies.

What is our research objective? Who are the results intended to apply to?

- A **methodologist** might be interested in understanding if the “correct” (best) version of their method is better than the “correct” version of a competitor's method...
- A **software developer** might want to know which version of a method is best across a range of scenarios, to understand whether to **change the default setting in Stata or R**
- An **applied researcher** might want to know how the default version of different methods work for their problem

Each of these questions requires a **different** simulation study

Returning to Slade and Naylor

- Previous simulation studies compared default versions of tree-based imputation methods vs. parametric methods
 - “Which methods are best for people like Brennan, who rely on default options?”
- Slade and Naylor compared the “optimised” versions of tree-based methods vs. parametric methods
 - “Which methods are best for people like Tim, who are experts in the methods?”

Which comparisons are “fair”?

- Depending on the aims (the inexpert vs. expert user), **both** might be fair
- ... or **not fair** (*i.e.* the **optimised comparison** wouldn't be fair for the **inexpert user**, while the **default comparison** wouldn't be fair for the **expert user**)
- The concept of “fair” depends on our objectives, and needs to be defined within that context
- There is no **objectively correct question** by which to define a “**fair**” comparison

Summary

- Identify your objectives (who are results intending to apply to? How would they use each method?)
- Design the simulation study in a fair way **based on these objectives** (e.g. if for the inexpert user, implement the inexpert methods)
- **Clarify** what your objectives are so others can tell whether you were fair

Part 1: planning (Tim)

- Introduction and overview
- Neutral comparisons?
- **Phases of methodological research**
- Elements of a simulation study: ADEMP



ORIGINAL ARTICLE

A method was developed for correcting the bias in the usual study weights in meta-analyses

Stephen D. Walter^{a,*}, Narayanaswamy Balakrishnan^b

^a*Department of Health Research Methods, Evidence, and Impact, McMaster University, Hamilton, Ontario, Canada*

^b*Department of Mathematics and Statistics, McMaster University, Hamilton, Ontario, Canada*

Accepted 29 August 2022; Published online 21 September 2022

Walter & Balakrishnan in brief

- The estimated (unscaled) weights in meta-analysis are biased upwards
- Find an expression for the bias and use this to derive a bias-correction
- Try it in two example meta-analyses with few, small studies; the weights are slightly different
- *‘We recommend that our bias correction should be routinely adopted’ and ‘Elimination of this bias will enhance the validity of conclusions from a meta-analysis, compared with the situation when the standard weights are used.’*



Let's give applied researchers credit

Applied researchers need to use a method that is suitable for their study, e.g. 'the particular trial I am designing', or 'this EHR dataset in front of me'.

Suppose they are motivated to do something as well as they can, rather than just to get something past journal editors / regulators (which always just amounts to precedence).

They might want to know **when a method is and is not preferred method; what diagnostics are available; pitfalls that may occur at application.**

LETTER TO THE EDITOR

Biometrical Journal →

On the necessity and design of studies comparing statistical methods

Anne-Laure Boulesteix¹ 

Harald Binder²

Michał Abrahamowicz³

Willi Sauerbrei²

for the Simulation Panel of the STRATOS Initiative

Argues strongly that we need to check our obsession with **novelty** and **innovation** and instead encourage ‘comparison studies’ as a valuable methodological endeavour.

Received: 12 August 2022

Revised: 9 December 2022

Accepted: 22 January 2023

DOI: 10.1002/bimj.202200222

Biometrical Journal **RESEARCH ARTICLE**

Phases of methodological research in biostatistics—Building the evidence base for new methods

Georg Heinze¹ | Anne-Laure Boulesteix² | Michael Kammer^{1,3} | Tim P. Morris⁴ | Ian R. White⁴ | on behalf of the Simulation Panel of the STRATOS initiative



Phases of drug development

PHASE I TRIALS: Initial studies to determine the metabolism and pharmacologic actions of drugs in humans, the side effects associated with increasing doses, and to gain early evidence of effectiveness; may include healthy participants and/or patients.

PHASE II TRIALS: Controlled clinical studies conducted to evaluate the effectiveness of the drug for a particular indication or indications in patients with the disease or condition under study and to determine the common short-term side effects and risks.

PHASE III TRIALS: Expanded controlled and uncontrolled trials after preliminary evidence suggesting effectiveness of the drug has been obtained and are intended to gather additional information to evaluate the overall benefit-risk relationship of the drug and provide an adequate basis for physician labelling.

PHASE IV TRIALS: Post-marketing studies to delineate additional information including the drug's risks, benefits, and optimal use.

Phases of methodological development

By analogy to drug development, Heinze *et al.* sketched a way to think about the development of statistical methods (some of you will have heard Georg present this at the previous session, if you picked well)

Key idea: what brings a method from ‘early phase’ to ‘late phase’, such that

1. We can be confident to recommend its use to applied researchers
2. Applied researchers can be confident in how to use & tune a method, to understand its limitations, pitfalls, etc.

After this phase, we know...

PHASE I: whether a method is valid – or not – from a theoretical point of view

PHASE II: whether a method can be used with caution or should not be used in certain applied settings

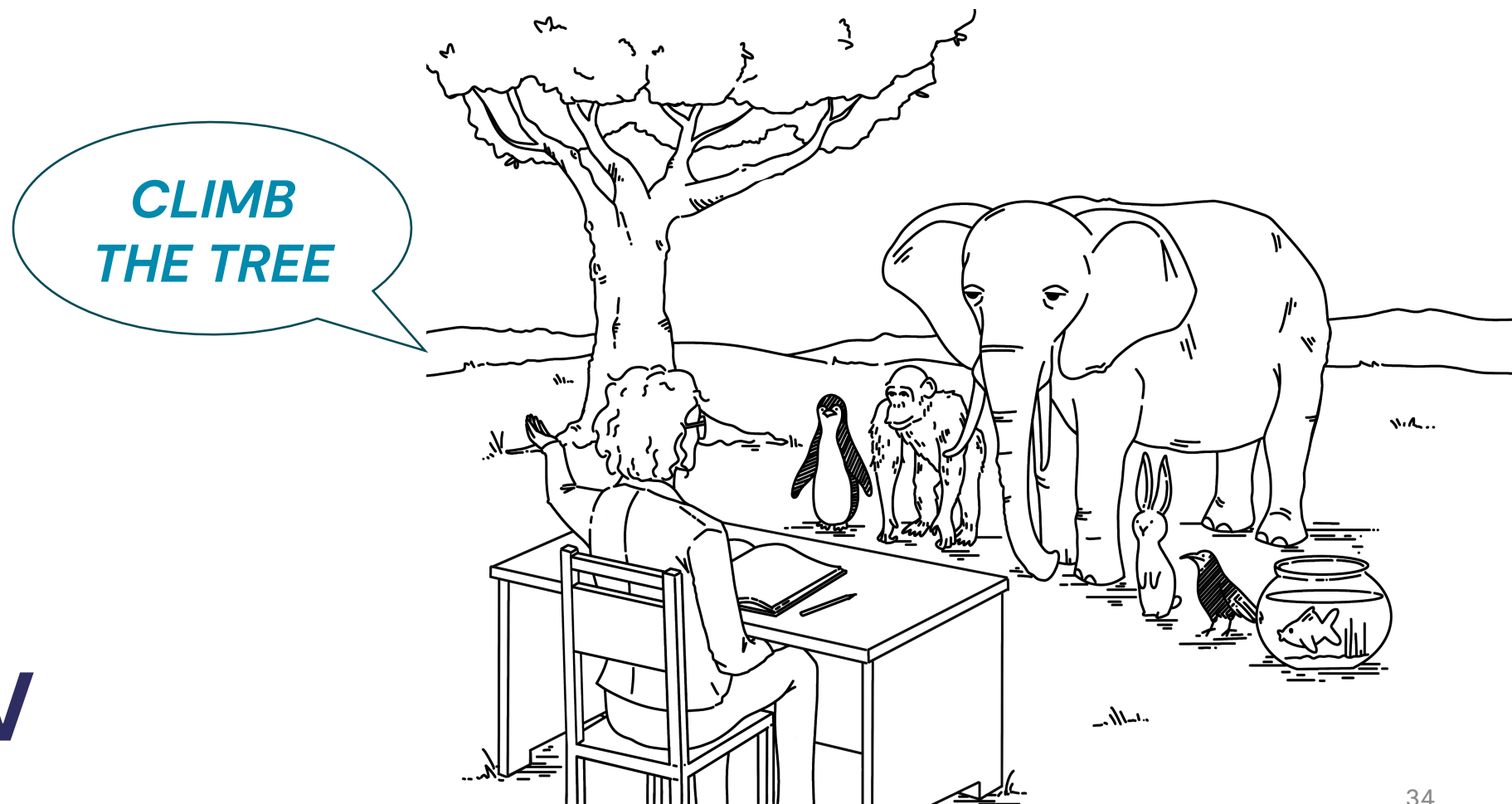
PHASE III: in which settings (among many) a method tends to outperform competing methods

PHASE IV TRIALS: when a method is and is not preferred; what diagnostics are available and which pitfalls may occur during its application

Against the “one method fits all data sets” philosophy for comparison studies in methodological research

Carolin Strobl Friedrich Leisch

First published: 02 September 2022 | <https://doi.org/10.1002/bimj.202200104> | Citations: 4



Phase III/IV

Phase I: derived correction and gave some simple examples

Biometrika (1993), 80, 1, pp. 27–38
Printed in Great Britain

Bias reduction of maximum likelihood estimates

BY DAVID FIRTH

Department of Mathematics, University of Southampton, SO9 5NH, U.K.

SUMMARY

It is shown how, in regular parametric problems, the first-order term is removed from the asymptotic bias of maximum likelihood estimates by a suitable modification of the score function. In exponential families with canonical parameterization the effect is to

Phase II: simulation study demonstrated that correction improved on existing methods

STATISTICS IN MEDICINE

Statist. Med. 2002; **21**:2409–2419 (DOI: 10.1002/sim.1047)

A solution to the problem of separation in logistic regression

Georg Heinze^{*,†} and Michael Schemper

*Section of Clinical Biometrics, Department of Medical Computer Sciences, University of Vienna,
Spitalgasse 23, A-1090 Vienna, Austria*

Phase III: comprehensive 'neutral' simulation study on logistic regression

van Smeden *et al.* *BMC Medical Research Methodology* (2016) 16:163
DOI 10.1186/s12874-016-0267-3

BMC Medical Research
Methodology

RESEARCH ARTICLE

Open Access

No rationale for 1 variable per 10 events criterion for binary logistic regression analysis



Maarten van Smeden^{1*} , Joris A. H. de Groot¹, Karel G. M. Moons¹, Gary S. Collins², Douglas G. Altman², Marinus J. C. Eijkemans¹ and Johannes B. Reitsma¹

Phase IV: explained why different methods lead to different results and gave advice on how to detect and deal with separation in practice



American Journal of Epidemiology

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Practice of Epidemiology

Separation in Logistic Regression: Causes, Consequences, and Control

Mohammad Ali Mansournia, Angelika Geroldinger*, Sander Greenland, and Georg Heinze

* Correspondence to Dr. Angelika Geroldinger, Section for Clinical Biometrics, Center for Medical Statistics, Informatics and Intelligent Systems, Medical University of Vienna, Austria Spitalgasse 23, 1090 Vienna, Austria (e-mail: angelika.geroldinger@gmx.at).

What phase?

What defines a ‘phase’ is an open discussion. We think the idea is helpful, despite the difficulty of pinning it down.

Slade & Naylor’s paper was a contribution to phase II/III:

1. Parametric imputation can perform as well – or better than – tree-based imputation in this setting, **if carefully implemented!**
2. The imputation method they use isn’t the best but is readily implemented in existing software, and idea was already well understood.

It’s not very broad and doesn’t mention pitfalls

Part 1: planning (Brennan)

- Introduction and overview
- Neutral comparisons?
- Phases of methodological research
- **Elements of a simulation study: ADEMP**

The core elements of a simulation study

ADEMP

Aim(s)

Data-generating mechanism(s)

Estimand(s)

Method(s)

Performance measure(s)

Think through these aspects before starting and your life will be easier

ADEMP: Aims

Set out what question(s) the simulation study is designed to address

ADEMP: Aims

Do we want to:

- Show parametric imputation *can be as good* as tree-based methods?
- Determine whether the *default settings* for parametric imputation should be changed?
- Evaluate which method is *better* across common settings?
- Or determine which is *more robust* to model misspecification?

ADEMP: Aims

In general, before we start we need to:

- Know what we want to learn
- Work out how we can learn it

The aims will inform **all** aspects of the design

ADEMP: Aims

Aim: show parametric imputation *fails* when it doesn't account for interactions

- A single **large** dataset **evaluating bias** may be enough

Aim: show parametric imputation *can be as good* tree-based methods?

- Need **multiple** simulated datasets, evaluating **bias** and **coverage**, but only under restricted data-generating mechanisms

Aim: Evaluate which method is *better* across common settings?

- Need **extensive simulation studies** under a wide range of data-generating mechanisms, evaluating **bias/precision/coverage/etc**

ADEMP: Data-generating mechanism (DGM)

Set out how the simulated data will be created.

Most simulation studies have multiple DGMs.

Follows on from [Aims](#).

ADEMP: Data-generating mechanism (DGM)

Aim: show parametric imputation *can be as good* tree-based methods

- Need a DGM where parametric imputation has been previously shown to fail
- Something with **strong** interactions between covariates

Aim: Evaluate which method is *better* across common settings?

- Many DGMs, both with and without interactions, and varying other factors (e.g. sample size, percentage missing data, pattern of missingness, etc)
- Aim is to have DGMs to cover most common settings

ADEMP: Data-generating mechanism

How do we generate data?

Two broad choices:

1. simulate data from a known model (**parametric simulation**)
2. **resample** with replacement from a set of data

Parametric simulation easier to adapt across a wide range of DGMs

Resampling naturally induces a realistic and complex DGM

Plasmode simulation uses elements of both

ADEMP: Data-generating mechanism

Should DGM factors be varied **factorially** or **one at a time**?

‘No aphorism is more frequently repeated in connection with field trials, than that we must ask Nature few questions, or, ideally, one question, at a time. The writer is convinced that this view is wholly mistaken.’

Fisher, 1926, The arrangement of field experiments

- If opting for one-at-a-time, can you rule out interactions?
- Full factorial simulations can be bulky → consequences for reporting
- Might go factorial for key factors and one-at-a-time for others
- Or fractional-factorial (see Skrondal, 2000)

ADEMP: Data-generating mechanism

Choice of DGMs should **also** be determined by choice of methods to compare

E.g. parametric vs. tree-based imputation:

- We know parametric methods are *usually* more precise, but *sometimes* at the expense of being less robust
- If we compared parametric vs. tree-based methods, but **didn't** include a DGM which included an interaction this would be **unfair**

ADEMP: Estimand/s

Before setting out how the simulated data will be analysed, we set out what quantity the analysis is trying to estimate

ADEMP: Estimand/s

Estimand = the thing we are trying to estimate
Sometimes called 'target of inference'

Important to understand

- The aims of analysis in practice
- Whether the analysis targets the estimand

Note:

Estimator = method for estimating an estimand

Estimate = number output by applying estimator

Estimands

(Credit: Simon Grund)



Ingredients

150g unsalted butter, plus extra for greasing

150g plain chocolate, broken into pieces

150g plain flour

½ tsp baking powder

½ tsp bicarbonate of soda

200g light muscovado sugar

2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.

estimand

estimate

estimator

ADEMP: Estimand/s

In epidemiology/RCTs, typically interested in **causal effect** of treatment/exposure

Statistical models often include treatment/exposure + other covariates (e.g. confounders in observational studies, baseline covariates in RCTs)

We're **not** interested in effect of covariates; our estimand is about the effect of **treatment/exposure**

ADEMP: Estimand/s

Not always so simple

E.g. we might be interested in estimating the ‘**shape**’ of exposure–outcome relationships

- This is awkward. How do we compare parameters for categorisation vs. splines vs. conventional polynomials vs. fractional polynomials?

Might estimate a contrast at $x=1$ vs. $x=0$ and compare to truth.

In a prognostic modelling study, we are not interested in the coefficients themselves, but in overall model performance.

What is the estimand? (calibration? Discrimination?)

ADEMP: Estimand/s

Methods targeting different estimands should only be compared with great care (if at all).

‘Intention to treat’ analysis targets the effect of being assigned a treatment (regardless of adherence etc.)

Some other analyses target the effect of adhering to treatment

Each (may) provide an unbiased estimate of the thing it claims to estimate

Each provides a biased estimate of the *other* estimand

ADEMP: Method/s

Set out how each simulated data set will be analysed to give an estimate of the estimand

ADEMP: Method/s

‘Method’ is quite generic.

E.g. in Slade and Naylor, could assess bias of the following imputation **procedure**:

Test for interaction between treatments in complete cases;

- If $p < \alpha$, do tree-based imputations;
- If $p \geq \alpha$, do parametric imputations;

ADEMP: Method/s

Methods should follow from our **Aims**

Aim: show parametric imputation *can be as good* tree-based methods

- Need to include both parametric and tree-based imputation

Aim: Determine whether the *default settings* for parametric imputation should be changed

- Only need to use parametric imputation...
- ... *but* need to include various implementations of parametric imputation

In general, it's usually good to have a comparator(s) method, but not always necessary

Can be good to include a familiar but dubious 'benchmark' method, e.g. complete cases in missing data work, as a check

Part 1: planning (Tim)

- Introduction and overview
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- **Elements of a simulation study: ADEMP**

ADEMP: Performance measures

What is a good result? Let's call the chosen estimand θ

Typically, a statistical method might output an estimate $\hat{\theta}$, its standard error $\widehat{se}(\hat{\theta})$ and a confidence interval $(\hat{\theta}_{low}, \hat{\theta}_{upp})$

- Often confidence interval is $\hat{\theta} \pm 1.96\widehat{se}(\hat{\theta})$ (for 95%)
- Might use t (if so, need to record degrees of freedom)
- Or another method – hard to cover everything here

We might be exploring a method that just outputs a p-value, or some measure of risk-prediction performance

ADEMP: Performance measures

The following performance measures may be of interest

Properties of estimator $\hat{\theta}$

- Bias
- Empirical SE
- MSE

Properties of SE estimator $\widehat{se}(\hat{\theta})$

- Average model-based SE

Properties of confidence interval $(\hat{\theta}_{low}, \hat{\theta}_{upp})$

- Coverage
- Power

Convergence

Computational speed

ADEMP: Performance measures

I'll talk through a few performance measures. Note:

- These measures relate to each other, sometimes in complex ways
- There is no 'correct' set to us, and this will depend on your aims, but a single performance measure will rarely suffice
- Definitions and concepts here; how we estimate performance will be later

Performance measures: Bias

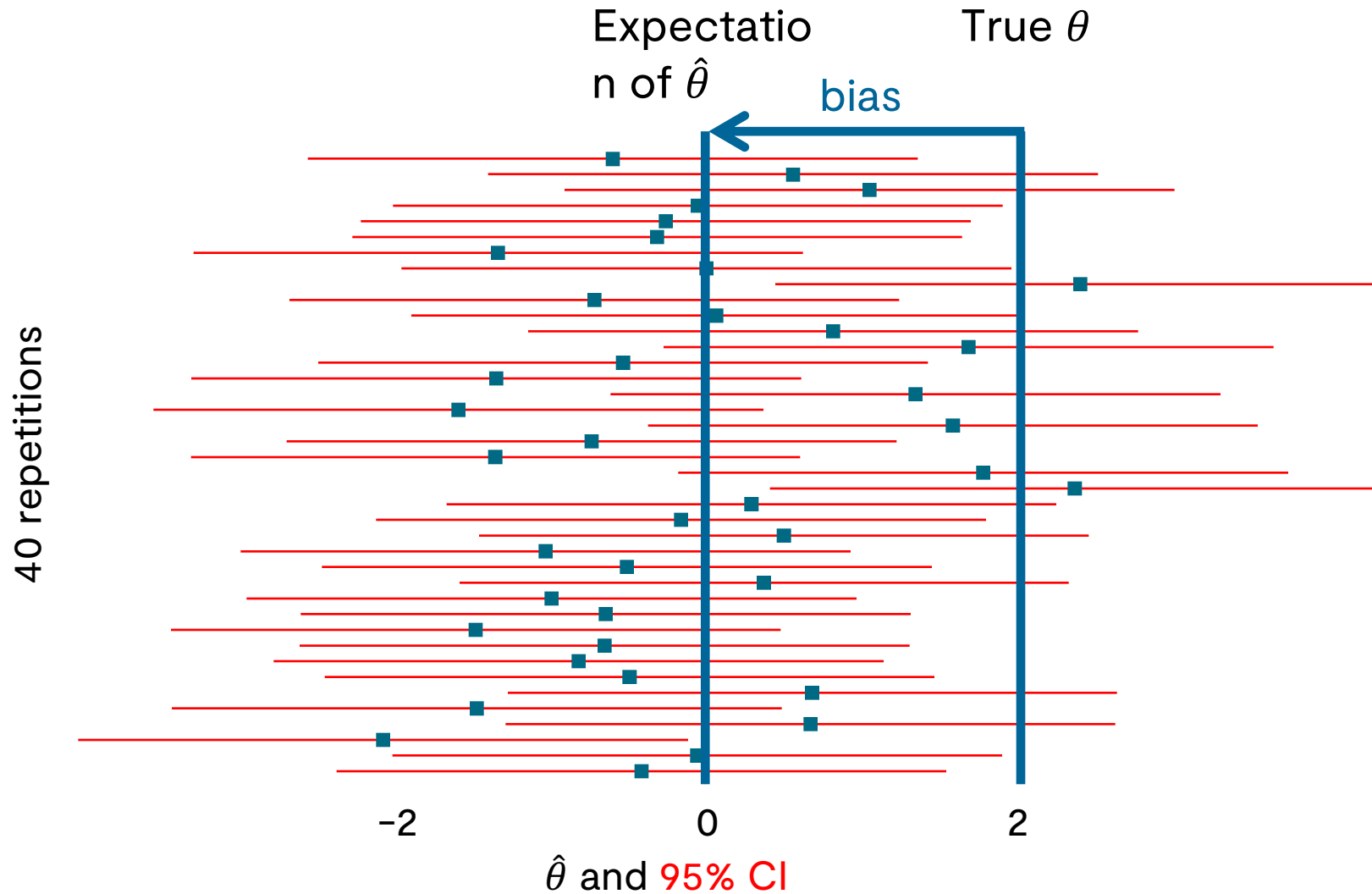
Unbiasedness is often regarded as a key property of an estimator, defined as $E[\hat{\theta}] - \theta = 0$

By how much does $\hat{\theta}$ exceed θ on average?

Two misnomers:

1. 'Average bias' is tautological and confusing
2. 'Mean absolute bias' is $E[|\hat{\theta}|] - \theta$ and this is not bias (it's an expected discrepancy or deviation from θ)

Performance measures: Bias



Performance measures: Empirical standard error / variance

We want precise estimators ('low variance'; 'efficient')

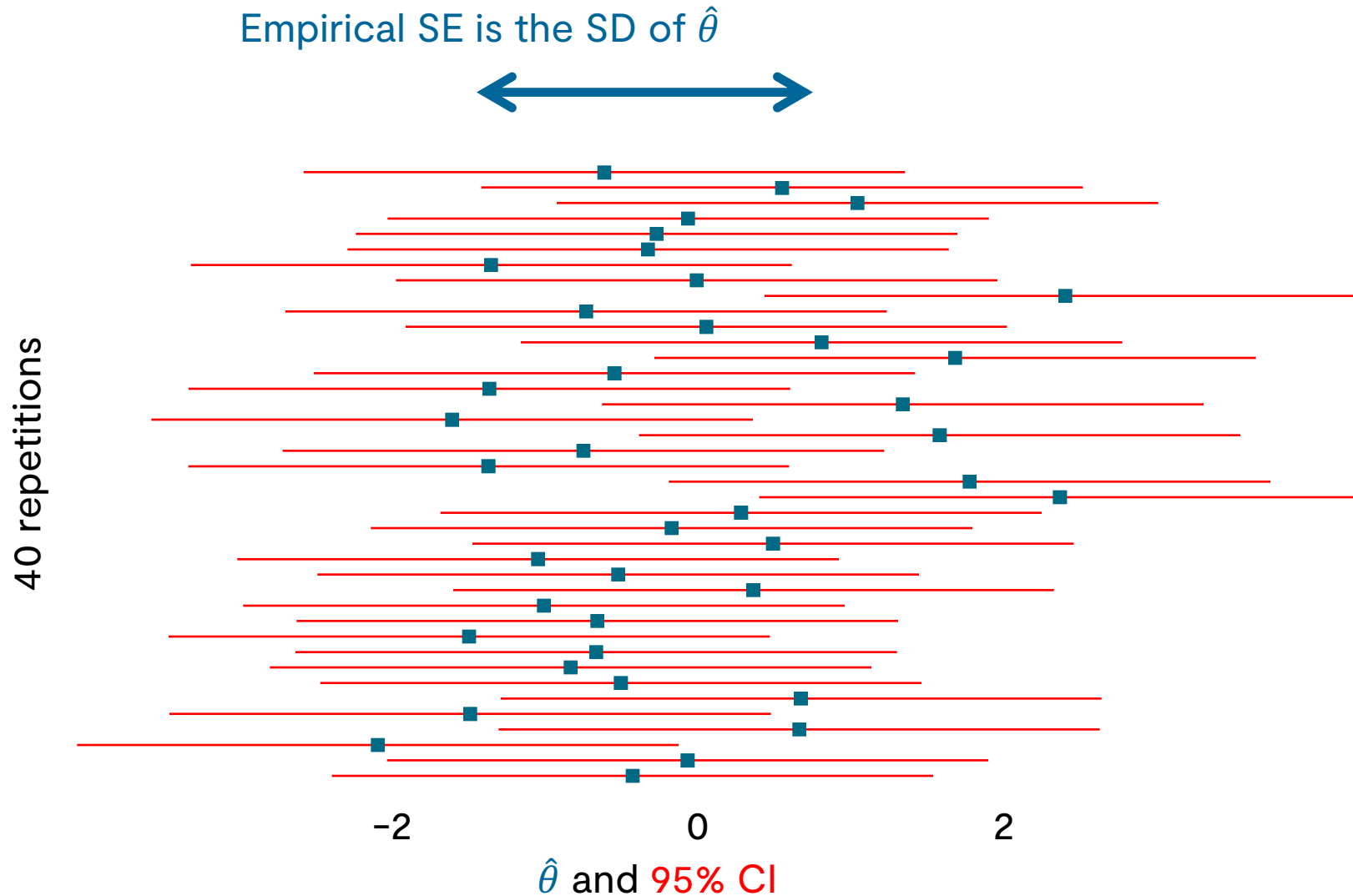
The empirical standard error is defined as

$$\text{EmpSE} = \sqrt{\text{Var}(\hat{\theta})}$$

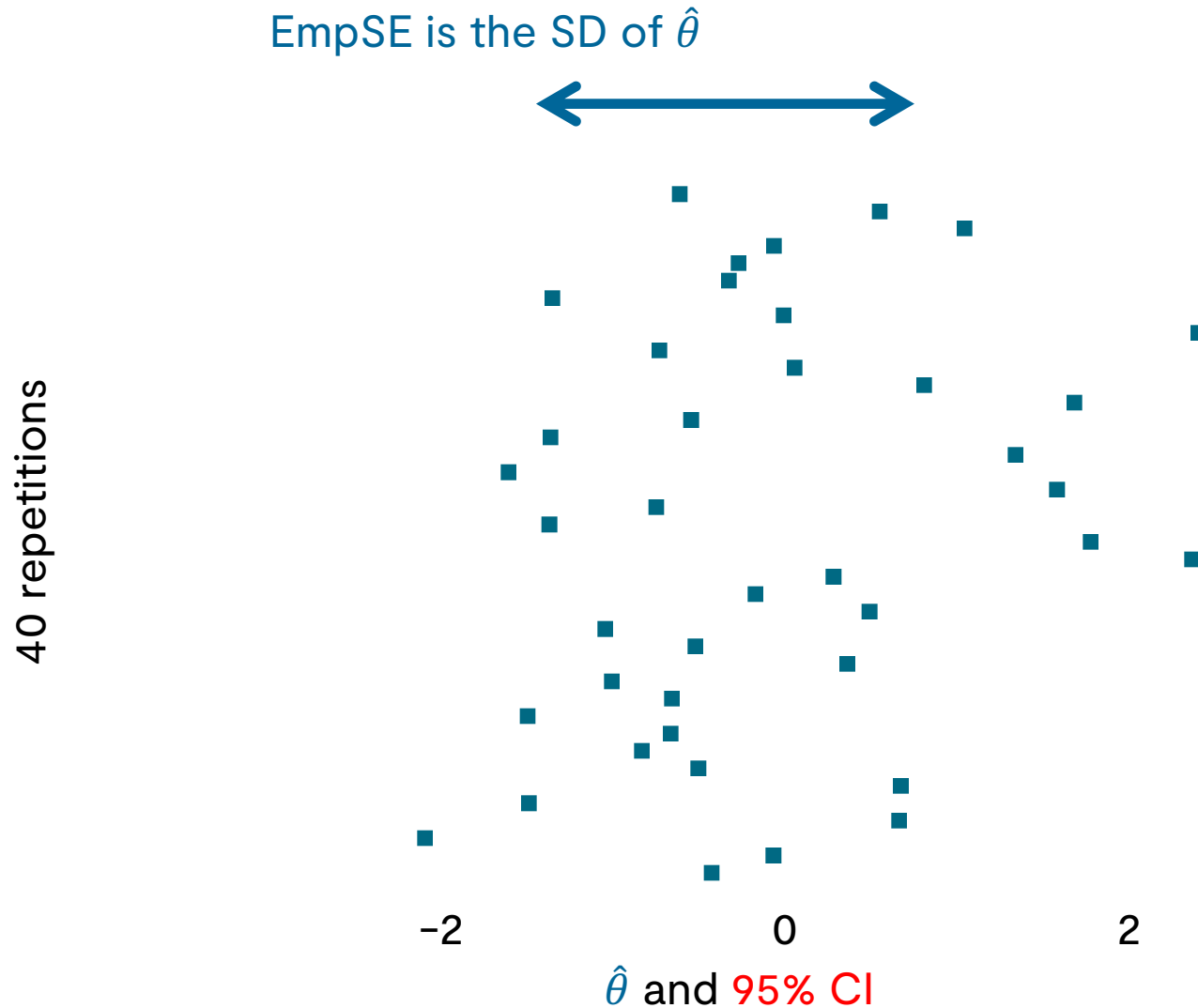
and should be as small as possible

- 'Empirical' is used to distinguish it from the standard error output by the analysis method
- Sometimes precision = $1/\text{Var}(\hat{\theta})$
- relative precision of $\hat{\theta}_2$ compared with $\hat{\theta}_1$ is $\text{Var}(\hat{\theta}_1)/\text{Var}(\hat{\theta}_2)$

Performance measures: Empirical standard error / Precision



Performance measures: Empirical standard error / Precision



Note on interpreting performance measures together

We will return to the broad point later. Please bear it in mind.

Empirical SE may not be meaningful for biased estimators

- because they often have smaller SE than unbiased estimators
- e.g. $\hat{\theta}/2$ must have greater precision than $\hat{\theta}$

Interpret cautiously: it is not the case that ‘well $\theta/2$ is biased but it is more precise so the two balance out’ in this case they have been exactly traded (in other cases the trade may be worthwhile)

Performance measures:

Mean squared error

Most natural **overall** performance measure for performance of $\hat{\theta}$?

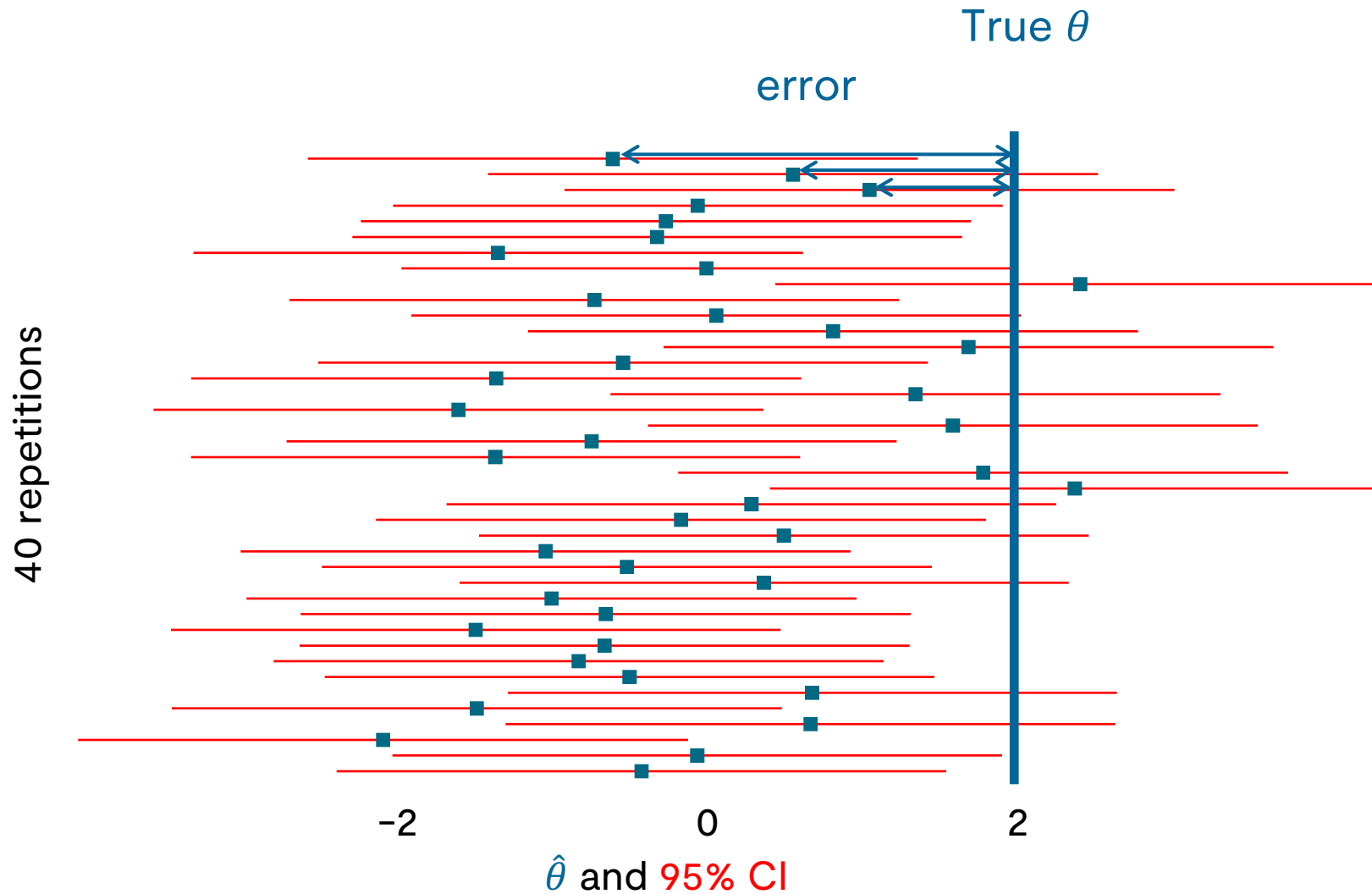
$$\text{MSE} = \text{E} \left[(\hat{\theta} - \theta)^2 \right]$$

MSE combines bias and empirical standard error:

$$\text{MSE} \approx \{ \text{E}[\hat{\theta} - \theta] \}^2 + \text{Var}(\hat{\theta}) = \text{Bias}^2 + \text{EmpSE}^2$$

Often easier to interpret **root mean squared error (RMSE)**

Performance measures: Mean squared error



Performance measures: average model-based SE

The empirical SE measures the spread of the estimator $\hat{\theta}$

But we also want to evaluate the standard error that comes with $\hat{\theta}$, i.e. $\widehat{se}(\hat{\theta})$

To focus ideas:

- Suppose we coded the standard error of the mean as SD/n instead of the correct formula $SD/\sqrt{n}$.
- How might a simulation study demonstrate our error?

Performance measures: average model-based SE

The squared standard error should be unbiased for the variance of the estimator

- e.g. the true variance of a Normal mean is σ^2/n and we estimate it as s^2/n where s^2 is an unbiased estimate of σ^2
- s is a biased estimate of σ

The average model-based SE is

$$\text{Avg. ModSE} = \sqrt{\text{E} \left[\widehat{\text{se}}(\hat{\theta})^2 \right]}$$

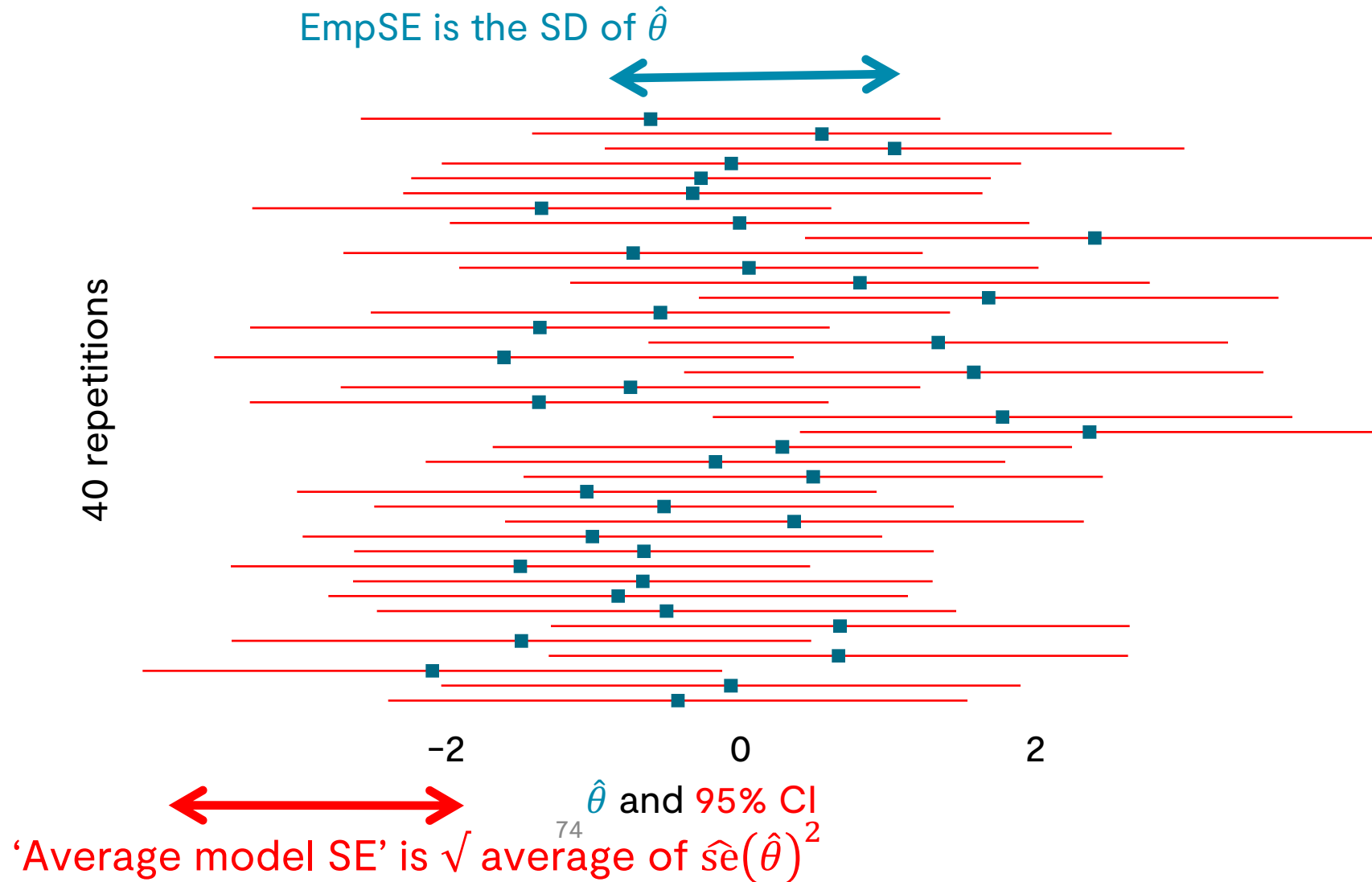
We want this to be equal to the true SE, represented by the empirical SE

Terminology is often unclear: be sure to distinguish EmpSE from ModSE

We can quantify the relative error in ModSE:

$$(\text{Mod. SE} / \text{Emp. SE}) - 1$$

Empirical SE and Average model SE



Performance measures:

Properties of confidence interval

Focus on 95% confidence intervals for definiteness

- Definition of a CI $(\hat{\theta}_{low}, \hat{\theta}_{upp})$:

$$P(\hat{\theta}_{low} \leq \theta \leq \hat{\theta}_{upp}) = 0.95$$

$$P(\theta \in (\hat{\theta}_{low}, \hat{\theta}_{upp})) = 0.95$$

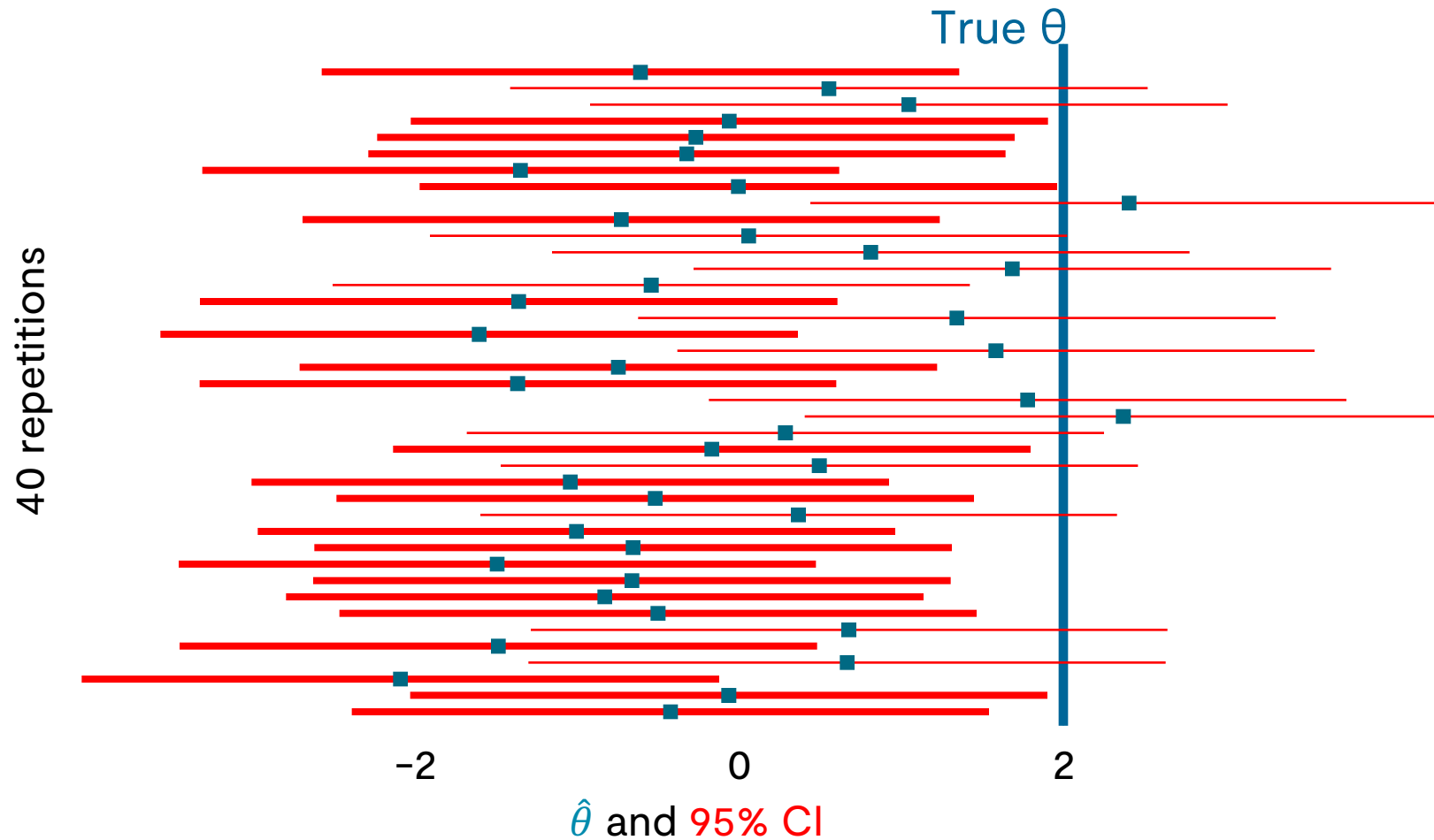
- Coverage is

$$\text{Cover} = P(\hat{\theta}_{low} \leq \theta \leq \hat{\theta}_{upp})$$

Poor coverage arises from any or all of

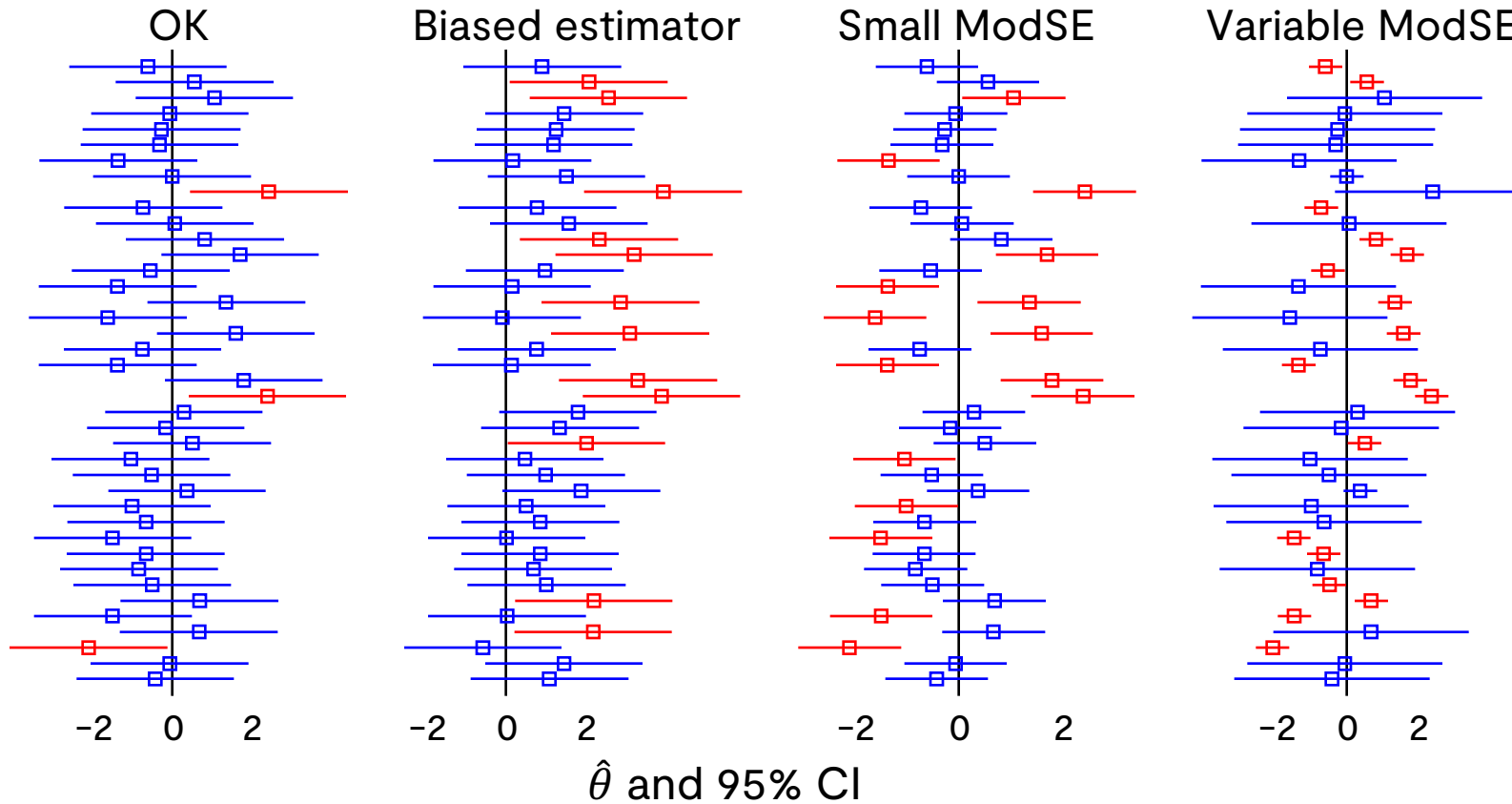
- bias in estimator
- bias in model SE
- poor distributional properties

Performance measures: Coverage



25 of 40 CIs (in bold) fail to include true $\theta \rightarrow$ estimated coverage is $15/40 = 37.5\%$

3 routes to low coverage



Coverage is ~95% for 'OK' and ~68% for others

Performance measures:

Type I error rate

Type I error rate is the probability of rejecting a null that is true

Particularly emphasised in clinical trials

Many tests amount to rejecting the null hypothesis whenever the null value is not in the confidence interval

Hence type I error is '1 – coverage' under the null

Performance measures: Power

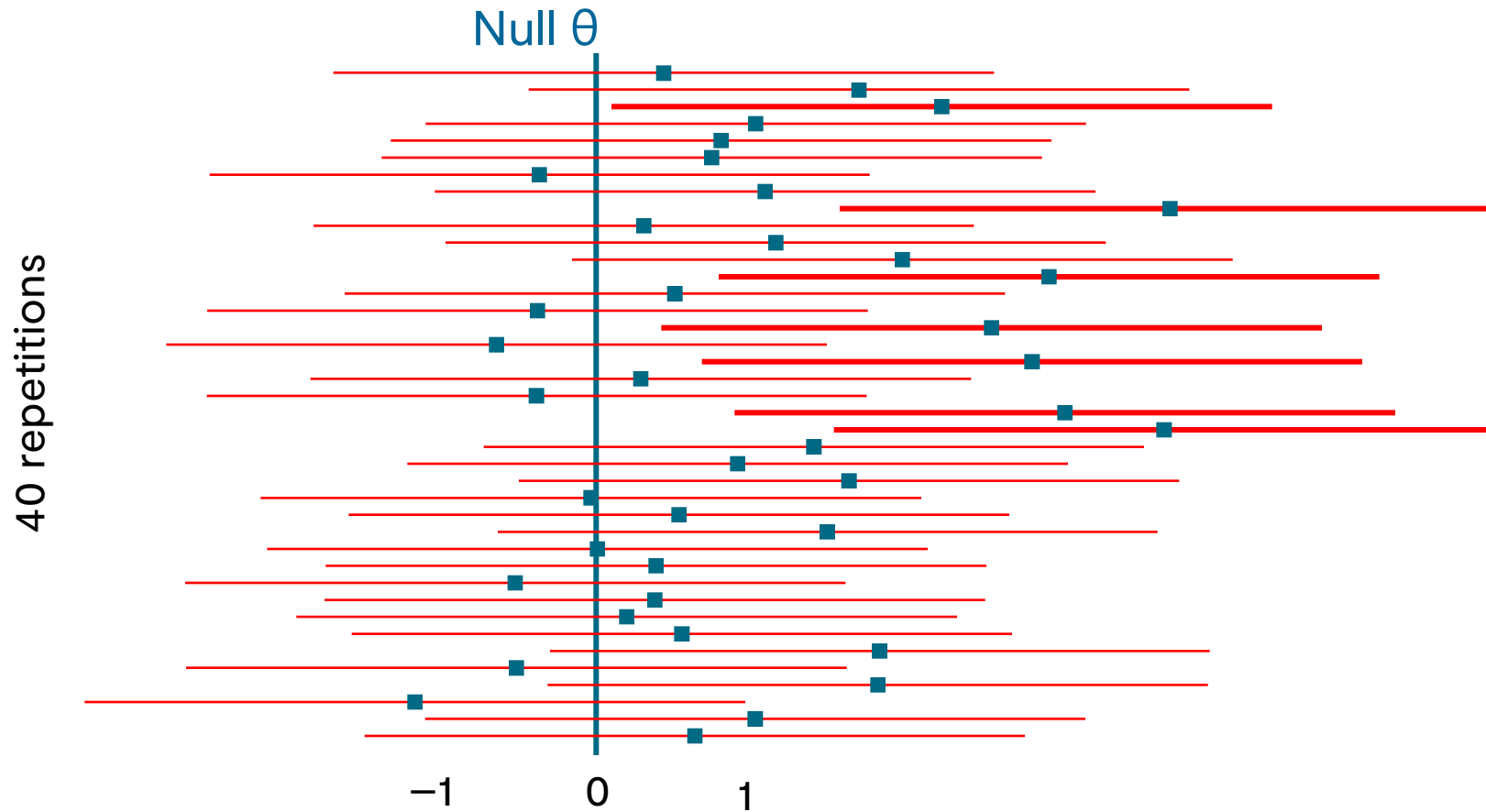
Defined as

$$\begin{aligned}\text{Power} &= P(0 \notin (\hat{\theta}_{low}, \hat{\theta}_{upp})) \\ &= P(\hat{\theta}_{low} > 0) + P(\hat{\theta}_{upp} < 0)\end{aligned}$$

– or one-sided version

- Especially of interest for study planning/design
- Blunt, but can be a way to compare biased estimators or estimators of different estimands, if they share a null
- Can handle other sorts of test, e.g. likelihood ratio test, permutation test; even tests that aren't associated with parameter estimates or CIs

Performance measures: Power



7 of 40 CIs (in bold) fail to include null $\theta \rightarrow$ estimated power is 7/40

Which performance measure/s?

- You wouldn't usually report performance for all the measures described above (sometimes you need different ones)
- Choice of performance measures should relate back to aims
- May have trade-offs between performance measures: e.g. bias might be accepted if it leads to models with better prognostic performance
- In general, need ≥ 3 performance measures capturing
 - is estimator biased? [Bias]
 - is estimator efficient? [EmpSE; Power]
 - is standard error adequate? [ModSE/EmpSE; Cov]

ADEMP

Aim(s)

Data-generating mechanism(s)

Estimand(s)

Method(s)

Performance measure(s)



MRC
Clinical
Trials Unit



Simulation studies as a tool to assess and compare the properties of statistical methods – an overview

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Part 2: coding, analysis, writing and reporting (Tim)

- **Coding a simulation study**
- Analysis of simulation studies
- Writing up and reporting
- Reproducibility?

Overview

This is not going to be detailed and will not include code but will discuss principles.

This is a big topic. Much of it relates to writing good code in general.

1. I will give a general approach to coding a simulation study
2. Later, Brennan will talk about checking results you don't believe and improving code to fix or reassure yourself

What does your code need to do?

1. Generate the required data
2. Analyse the generated data
3. Store results

Aims in coding a simulation study

1. Want to be confident that your simulation study was coded well enough to avoid errors, to check results you don't believe, and to find/fix issues
2. Need code to be structured so that someone else can read and adapt it
3. Want to store results in a well-structured 'estimates' dataset

Simulated datasets and Estimates dataset

rep_no	x1	x2	y
1	16.35013	1	38.37179
1	24.2343	0	34.14901

1	3.	rep_no	x1	x2	y
1	8.	2	23.86639	0	47.72385
1	12	2	26.77522	1	58.26255

1	23	2	rep_no	x1	x2	y
1	17	2	3	17.67529	1	49.35355
1	3.	2	3	10.84167	0	58.08726
1	22	2	3	9.639807	0	47.08185
1	7.	2	3	28.05374	1	56.96053
1	24	2	3	24.19091	1	50.6784
1	2	2	3	20.76223	1	64.59888
1	7.	2	3	27.16945	1	42.93995
1	18	2	3	26.54299	0	50.52908
1	15	2	3	23.86617	1	47.02399
1	21	2	3	11.20086	0	47.83489
1	19	2	3	12.41312	0	62.99809
1	9.	2	3	29.38236	1	51.41874
1	.9	2	3	14.01399	0	65.50615
1	23	2	3	10.74801	0	52.5156
		2	3	21.1476	1	55.21921
		2	3	16.47218	0	49.29565
		2	3	14.00056	1	54.68225
			3	16.78905	1	61.35783
			3	3.979811	0	44.87682
			3	2.89212	1	62.30411

Simulated
datasets

rep_no	theta	se
1	.7253602	.113077
2	.9061058	1.571631
3	-2.685881	.3420264
4	-1.277903	.4767973
5	-.5239586	.1492868
6	.8658702	.0418173
7	1.040118	.6079704
8	-1.568616	.4439402
9	.2409494	.3497852
10	-1.188984	.0971455
11	-.1022815	.0010391
12	-.6783351	.2035454
13	-2.006534	.0005921
14	-.3832266	.0661816
15	-1.33277	.0143398
16	-.5361734	.8849165
17	1.535977	.3798626
18	1.089039	.5508905
19	-1.446115	.1594974
20	-.6660534	.7300485

Estimates
dataset

A well-structured estimates dataset

	rep	DGM	Method	β_0	β_1	β_2	β_3	$se\beta_0$	$se\beta_1$	$se\beta_2$	$se\beta_3$
1	1	Scenario 1	PMM-Na...	.	.	.	.	.	.	.	.
2	1	Scenario 1	PMM-Int	.	.	.	.	.	.	.	.
3	1	Scenario 1	CART	.	.	.	.	.	.	.	.
4	1	Scenario 1	RF	.	.	.	.	.	.	.	.
5	1	Scenario 2	PMM-Na...	.	.	.	.	.	.	.	.
6	1	Scenario 2	PMM-Int	.	.	.	.	.	.	.	.
7	1	Scenario 2	CART	.	.	.	.	.	.	.	.
8	1	Scenario 2	RF	.	.	.	.	.	.	.	.
9	2	Scenario 1	PMM-Na...	.	.	.	.	.	.	.	.
10	2	Scenario 1	PMM-Int	.	.	.	.	.	.	.	.
11	2	Scenario 1	CART	.	.	.	.	.	.	.	.
12	2	Scenario 1	RF	.	.	.	.	.	.	.	.
13	2	Scenario 2	PMM-Na...	.	.	.	.	.	.	.	.
14	2	Scenario 2	PMM-Int	.	.	.	.	.	.	.	.
15	2	Scenario 2	CART	.	.	.	.	.	.	.	.
16	2	Scenario 2	RF	.	.	.	.	.	.	.	.

A variable for:

1. Repetition number
2. DGM
3. Method of analysis

So a row of data gives estimates from the given method under the given DGM for the given repetition. Not a great idea to have 'long' format betas!

Start small and build up slowly

Your code can – and likely will – go wrong. It would be naïve to imagine otherwise.

Make it easy to identify problems quickly with tests, and to and isolate them with suitably modular and modifiable code.

Example: for a simulation study involving adaptive designs, begin with no adaptations

Be paranoid: check everything that you can as you go

Unexpected things happen even for seemingly simple cases!

Example: I want $x \sim N(2, 4)$, then it's

`x <- rnorm(n=1, mean=0.2, sd=2)` in R

`gen x = rnormal(2, 2)` in Stata

I've seen several people get tripped-up by this because we're used to writing and reading $\sim N(\mu, \sigma^2)$

Get things running for a single simulated dataset

This is particularly useful for storing results. Verify that

1. The results you see are also the results stored in your **estimates dataset** (e.g. what scale are the parameter estimates on and is it the one you want)
2. The estimates dataset is well structured for one repetition and that results are being stored in the right place

When you build something on, check it works as expected

Building something on typically involves extending an existing DGM, adding a new DGM, or adding methods of analysis

Check what this does and, as before, ensure you are storing the right results and check the estimates dataset

Two common mistakes:

1. Copy–paste code to return/store results (accidentally saving the results of one analysis twice)
2. Changing a model and storing the wrong parameter (e.g. it was previously [1,2] but is now [1,3])

The ‘three tasks’

To the extent possible, it's a good idea to separate code for the three tasks. For simulation studies of any complexity, I tend to write:

1. One or more programs/functions to **generate a simulated dataset** (usually with arguments for different DGMs)
2. One or more programs/functions to **analyse the generated data** and returns results of interest
3. A script that **repeatedly calls** these programs/functions and **stores results** in a well-structured estimates dataset

Keeping these tasks as separate as possible helps to isolate the source of errors

Reproducing a single simulated dataset

After running n repetitions, we might want to understand what happened in one, and need to be able to look at it. There are two options here:

1. Alessandro Gasparini outputs every simulated dataset*, so he can simply open it
2. I sometimes store the RNG state before simulating the dataset. I can then return to this state and re-run my program/function that produced it.

*Someone recently tried to be tidier, and put them all into one large dataset with a repetition number, with disastrous consequences

Setting seeds

It's useful to set a seed to ensure we can replicate our own results

NEVER set the seed within your repetitions. Ok, I'll use some pseudo-code to explain what to do and what not to do.

GOOD

```
SET RandomSeed to #  
FOR Repetition 1 to n_sim by 1  
    GENERATE simulated dataset  
    ...  
END FOR
```

BAD

```
FOR Repetition 1 to n_sim by 1  
    SET RandomSeed to #  
    GENERATE simulated dataset  
    ...  
END FOR
```

Run three repetitions

I've sometimes seen packages where the author set the seed within the package.

- This is very bad, because it leads to the problem on the previous slide.
- When you think you are ready to run your full simulation study, **run three repetitions**.
- This can identify the fixed-seed problem because results for reps 2 and 3 are then identical.

Part 2: coding, analysis, writing and reporting (Tim)

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Simulated datasets and Estimates dataset

rep_no	x1	x2	y			
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1	.9	2	3	14.01399	0	65.50615
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6	.8658702	.0418173
7	1.040118	.6079704
8	-1.568616	.4439402
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Estimates
dataset

A well-structured estimates dataset

	rep	DGM	Method	β_0	β_1	β_2	β_3	$se\beta_0$	$se\beta_1$	$se\beta_2$	$se\beta_3$
1	1	Scenario 1	PMM-Na...	.	.	.	.	.	.	.	.
2	1	Scenario 1	PMM-Int	.	.	.	.	.	.	.	.
3	1	Scenario 1	CART	.	.	.	.	.	.	.	.
4	1	Scenario 1	RF	.	.	.	.	.	.	.	.
5	1	Scenario 2	PMM-Na...	.	.	.	.	.	.	.	.
6	1	Scenario 2	PMM-Int	.	.	.	.	.	.	.	.
7	1	Scenario 2	CART	.	.	.	.	.	.	.	.
8	1	Scenario 2	RF	.	.	.	.	.	.	.	.
9	2	Scenario 1	PMM-Na...	.	.	.	.	.	.	.	.
10	2	Scenario 1	PMM-Int	.	.	.	.	.	.	.	.
11	2	Scenario 1	CART	.	.	.	.	.	.	.	.
12	2	Scenario 1	RF	.	.	.	.	.	.	.	.
13	2	Scenario 2	PMM-Na...	.	.	.	.	.	.	.	.
14	2	Scenario 2	PMM-Int	.	.	.	.	.	.	.	.
15	2	Scenario 2	CART	.	.	.	.	.	.	.	.
16	2	Scenario 2	RF	.	.	.	.	.	.	.	.

Question: what is the correlation structure in this estimates dataset?

Estimates data analysis

Having run the simulation study, we are going to analyse the estimates dataset to estimate performance.

As with any analysis, we want to check and explore this dataset

- Make sure we know what the data look like and where they came from
- Check for outliers
- Know about missing values

Etc. In short, it's business as usual

Multiple imputation with interactions

I am going to mimic part of Slade & Naylor's simulation study with slightly different methods, to demonstrate ideas

In doing this, I discovered things... will return to reproducibility later

$n_{sim} = 300$ reps (more later); S&N used $n_{sim} = 10,000$

First thing: check for missing data!

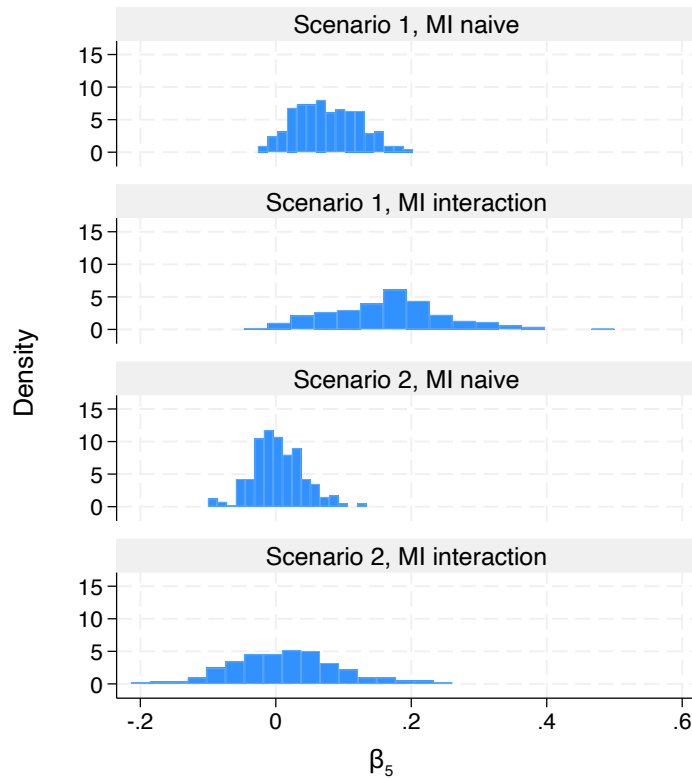
I get

1. 26/300 missing rows of results for scenario 1
2. 18/300 missing rows of results for scenario 2

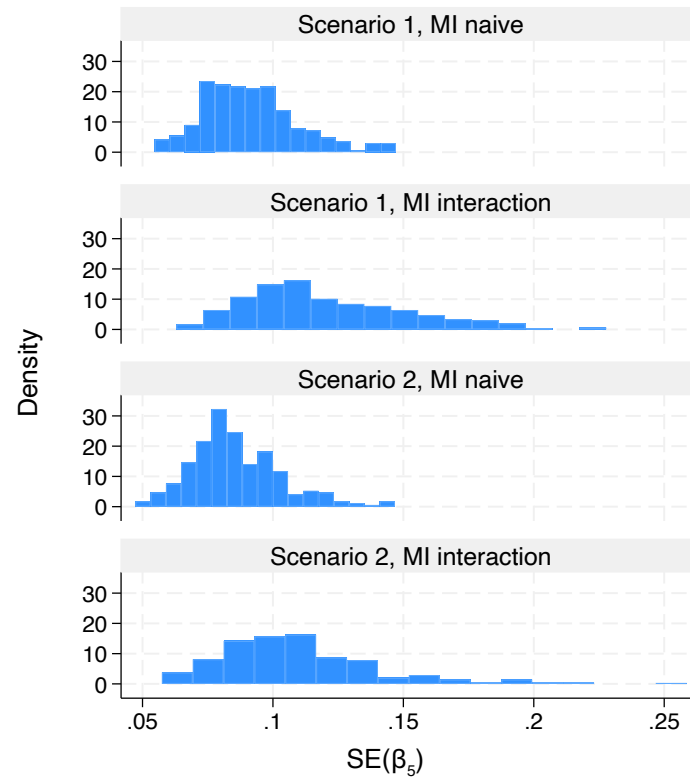
This turns out to be because the missingness mechanism sometimes produced an extremely low proportion of complete cases (e.g. 6 rows with all variables observed).

I have ideas on how to handle this but we would need another hour

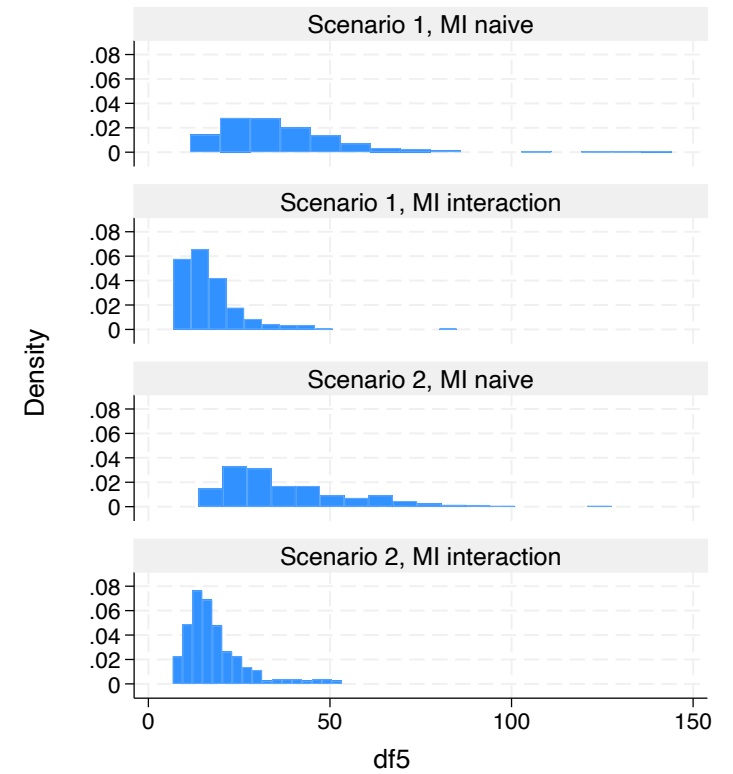
Plot the data!



Graphs by DGM and Method

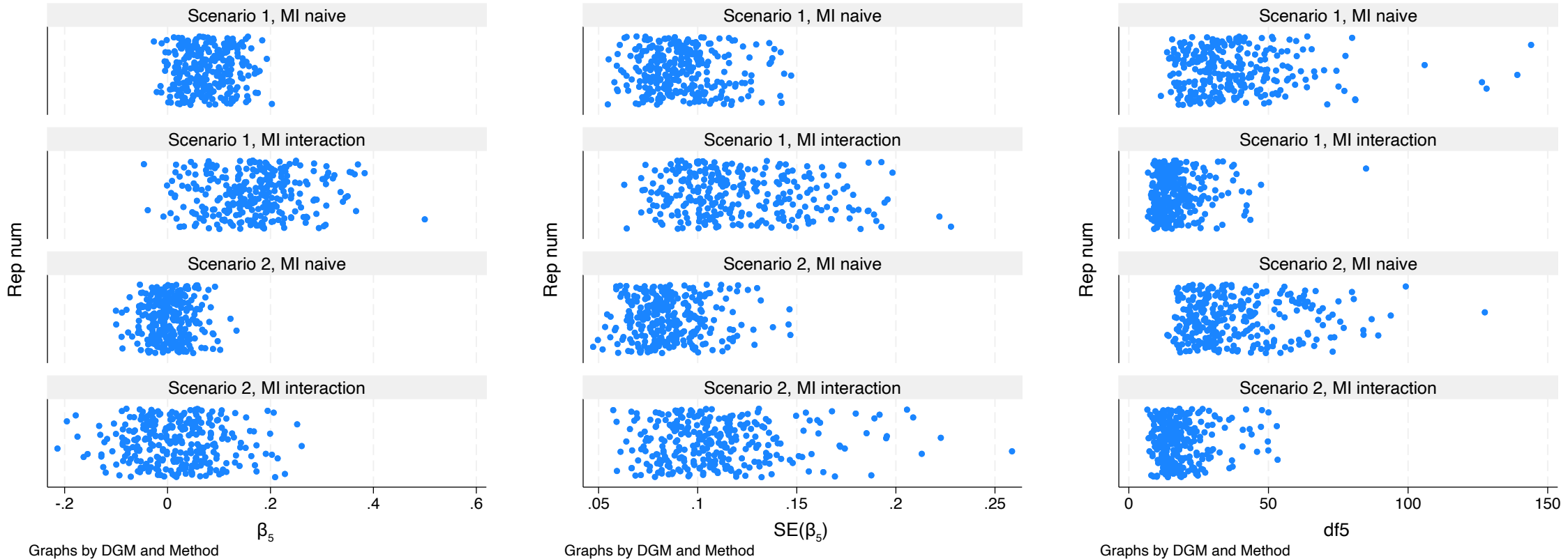


Graphs by DGM and Method



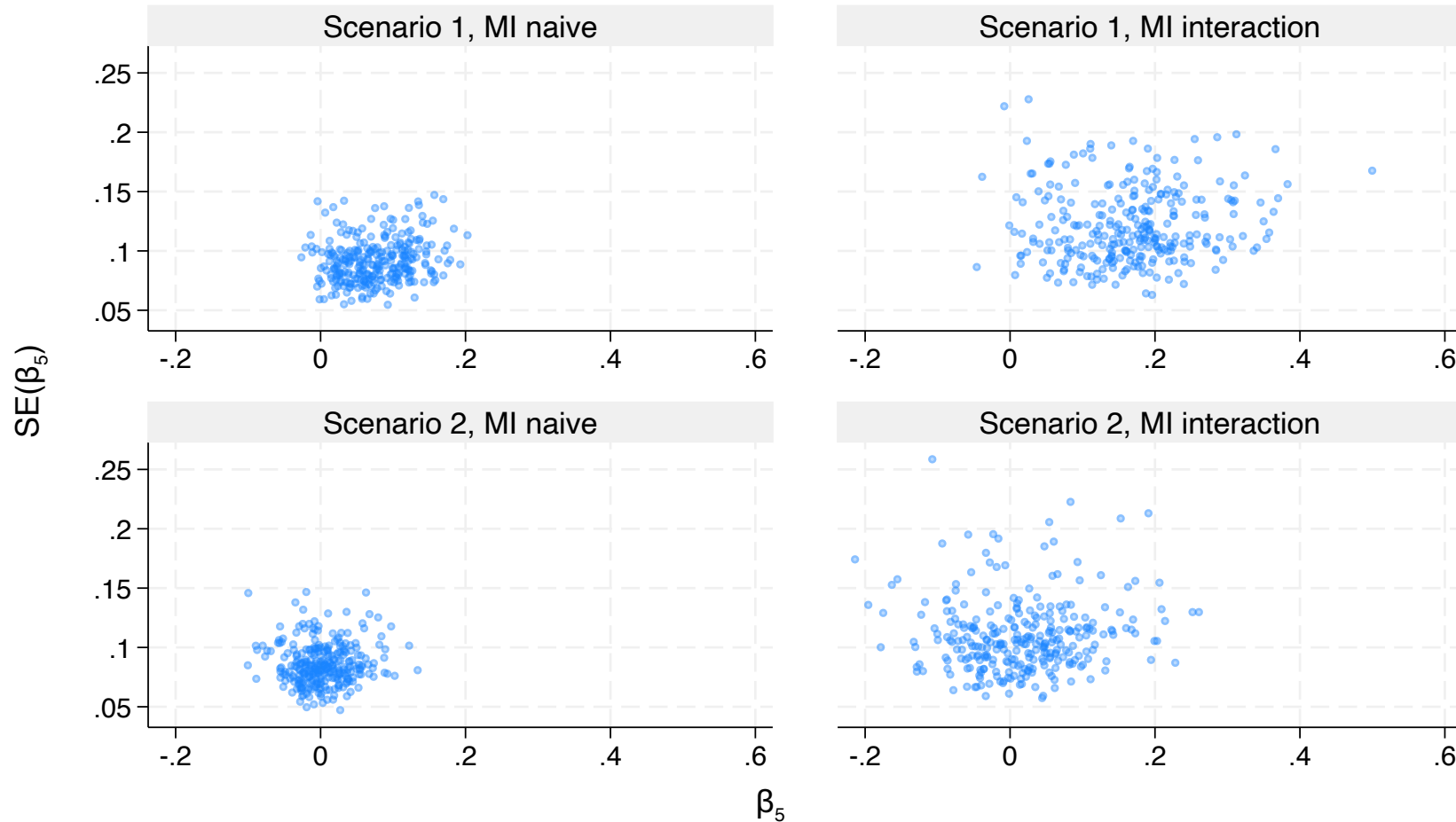
Graphs by DGM and Method

Useful check for outliers & code errors: repetition number vs. estimate



Estimate vs. SE estimate

Check for bivariate outliers



Graphs by DGM and Method

We are now ready to estimate performance

Estimation of performance

Earlier on, I gave you some **definitions** of performance measures

We have **estimators** of performance (which are fairly obvious from the definitions)

Since we are good statisticians, we also want to acknowledge uncertainty about this, which we do using MCSE (the '**Monte Carlo Standard Error**')

What / use the tutorial paper for:

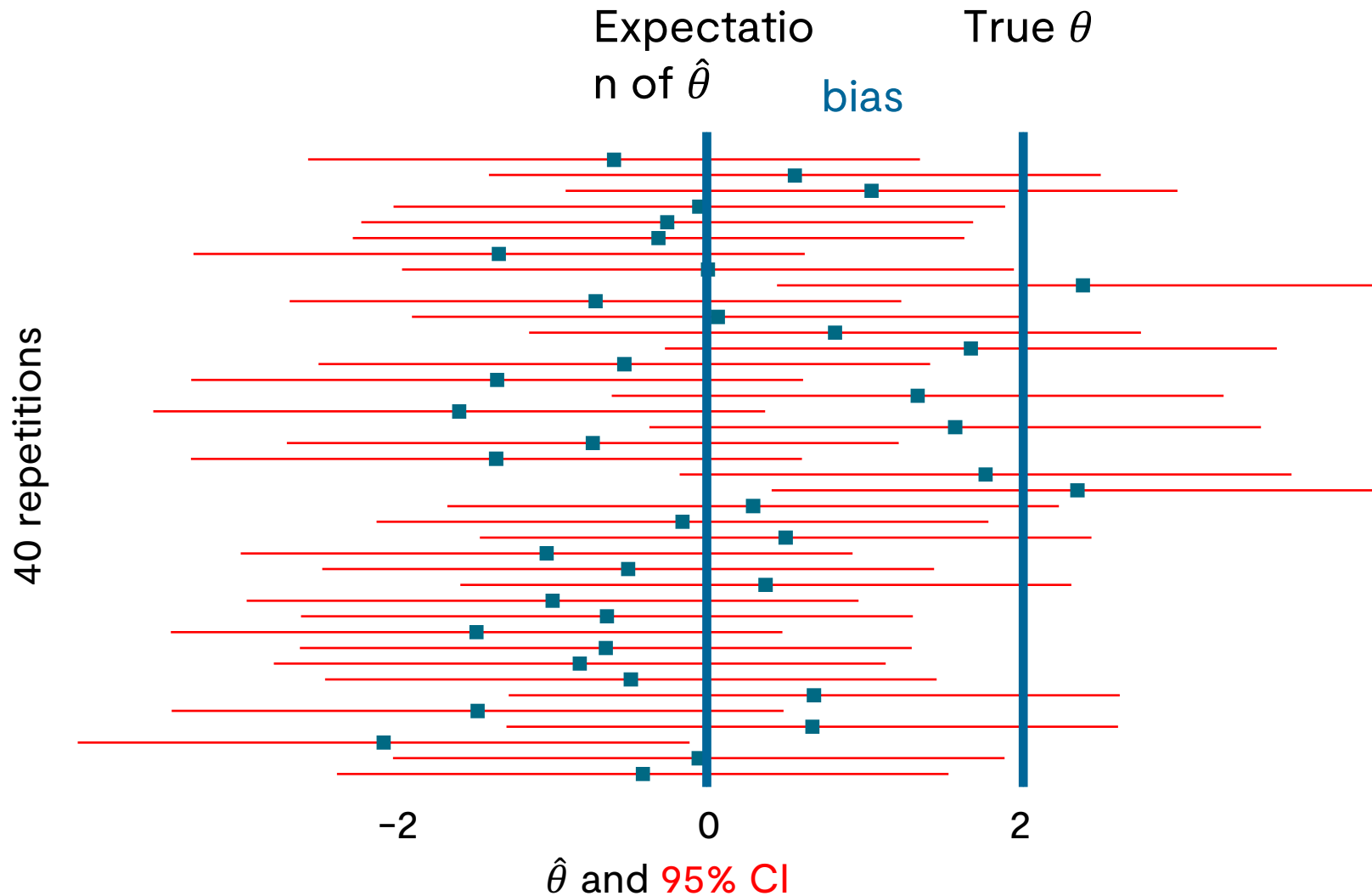
TABLE 6 Performance measures: definitions, estimates and Monte Carlo standard errors

Performance Measure	Definition	Estimate	Monte Carlo SE of Estimate
Bias	$E[\hat{\theta}] - \theta$	$\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i - \theta$	$\sqrt{\frac{1}{n_{\text{sim}}(n_{\text{sim}}-1)} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \bar{\theta})^2}$
EmpSE	$\sqrt{\widehat{\text{Var}}(\hat{\theta})}$	$\sqrt{\frac{1}{n_{\text{sim}}-1} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \bar{\theta})^2}$	$\frac{\widehat{\text{EmpSE}}}{\sqrt{2(n_{\text{sim}}-1)}}$
Relative % increase in precision (B vs A) ^a	$100 \left(\frac{\widehat{\text{Var}}(\hat{\theta}_A)}{\widehat{\text{Var}}(\hat{\theta}_B)} - 1 \right)$	$100 \left(\left(\frac{\widehat{\text{EmpSE}}_A}{\widehat{\text{EmpSE}}_B} \right)^2 - 1 \right)$	$200 \left(\frac{\widehat{\text{EmpSE}}_A}{\widehat{\text{EmpSE}}_B} \right)^2 \sqrt{\frac{1 - \text{Corr}(\hat{\theta}_A, \hat{\theta}_B)^2}{n_{\text{sim}} - 1}}$
MSE	$E[(\hat{\theta} - \theta)^2]$	$\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta)^2$	$\sqrt{\frac{\sum_{i=1}^{n_{\text{sim}}} [(\hat{\theta}_i - \theta)^2 - \widehat{\text{MSE}}]^2}{n_{\text{sim}}(n_{\text{sim}}-1)}}$
Average ModSE ^a	$\sqrt{E[\widehat{\text{Var}}(\hat{\theta})]}$	$\sqrt{\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \widehat{\text{Var}}(\hat{\theta}_i)}$	$\sqrt{\frac{\widehat{\text{Var}}[\widehat{\text{Var}}(\hat{\theta})]}{4n_{\text{sim}} \times \widehat{\text{ModSE}}^2}} \quad \text{b}$
Relative % error in ModSE ^a	$100 \left(\frac{\widehat{\text{ModSE}}}{\widehat{\text{EmpSE}}} - 1 \right)$	$100 \left(\frac{\widehat{\text{ModSE}}}{\widehat{\text{EmpSE}}} - 1 \right)$	$100 \left(\frac{\widehat{\text{ModSE}}}{\widehat{\text{EmpSE}}} \right) \sqrt{\frac{\widehat{\text{Var}}[\widehat{\text{Var}}(\hat{\theta})]}{4n_{\text{sim}} \times \widehat{\text{ModSE}}^4} + \frac{1}{2(n-1)}} \quad \text{b}$
Coverage	$\Pr(\hat{\theta}_{\text{low}} \leq \theta \leq \hat{\theta}_{\text{upp}})$	$\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} 1(\hat{\theta}_{\text{low},i} \leq \theta \leq \hat{\theta}_{\text{upp},i})$	$\sqrt{\frac{\widehat{\text{Cover}} \times (1 - \widehat{\text{Cover}})}{n_{\text{sim}}}}$
Bias-eliminated coverage	$\Pr(\hat{\theta}_{\text{low}} \leq \bar{\theta} \leq \hat{\theta}_{\text{upp}})$	$\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} 1(\hat{\theta}_{\text{low},i} \leq \bar{\theta} \leq \hat{\theta}_{\text{upp},i})$	$\sqrt{\frac{\text{B-E } \widehat{\text{Cover}} \times (1 - \text{B-E } \widehat{\text{Cover}})}{n_{\text{sim}}}}$
Rejection % (power or type I error)	$\Pr(p_i \leq \alpha)$	$\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} 1(p_i \leq \alpha)$	$\sqrt{\frac{\widehat{\text{Power}} \times (1 - \widehat{\text{Power}})}{n_{\text{sim}}}}$

^aMonte Carlo SEs are approximate for *Relative % increase in precision*, *Average ModSE*, and *Relative % error in ModSE*.

^b $\widehat{\text{Var}}[\widehat{\text{Var}}(\hat{\theta})] = \frac{1}{n_{\text{sim}}-1} \sum_{i=1}^{n_{\text{sim}}} \{\widehat{\text{Var}}(\hat{\theta}_i) - \frac{1}{n_{\text{sim}}} \sum_{j=1}^{n_{\text{sim}}} \widehat{\text{Var}}(\hat{\theta}_j)\}^2$.

Bias



True bias:

$$\text{Bias} = E[\hat{\theta}] - \theta$$

Estimator of bias:

$$\widehat{\text{Bias}} = \frac{1}{n_{\text{sims}}} \sum_{i=1}^{n_{\text{sims}}} \hat{\theta} - \theta$$

This is just a sample mean (minus a constant)

Consider whether to report bias or % bias?

MCSE of bias

Estimator of bias: $\widehat{\text{Bias}} = \frac{1}{n_{\text{sims}}} \sum_{i=1}^{n_{\text{sims}}} \hat{\theta} - \theta$

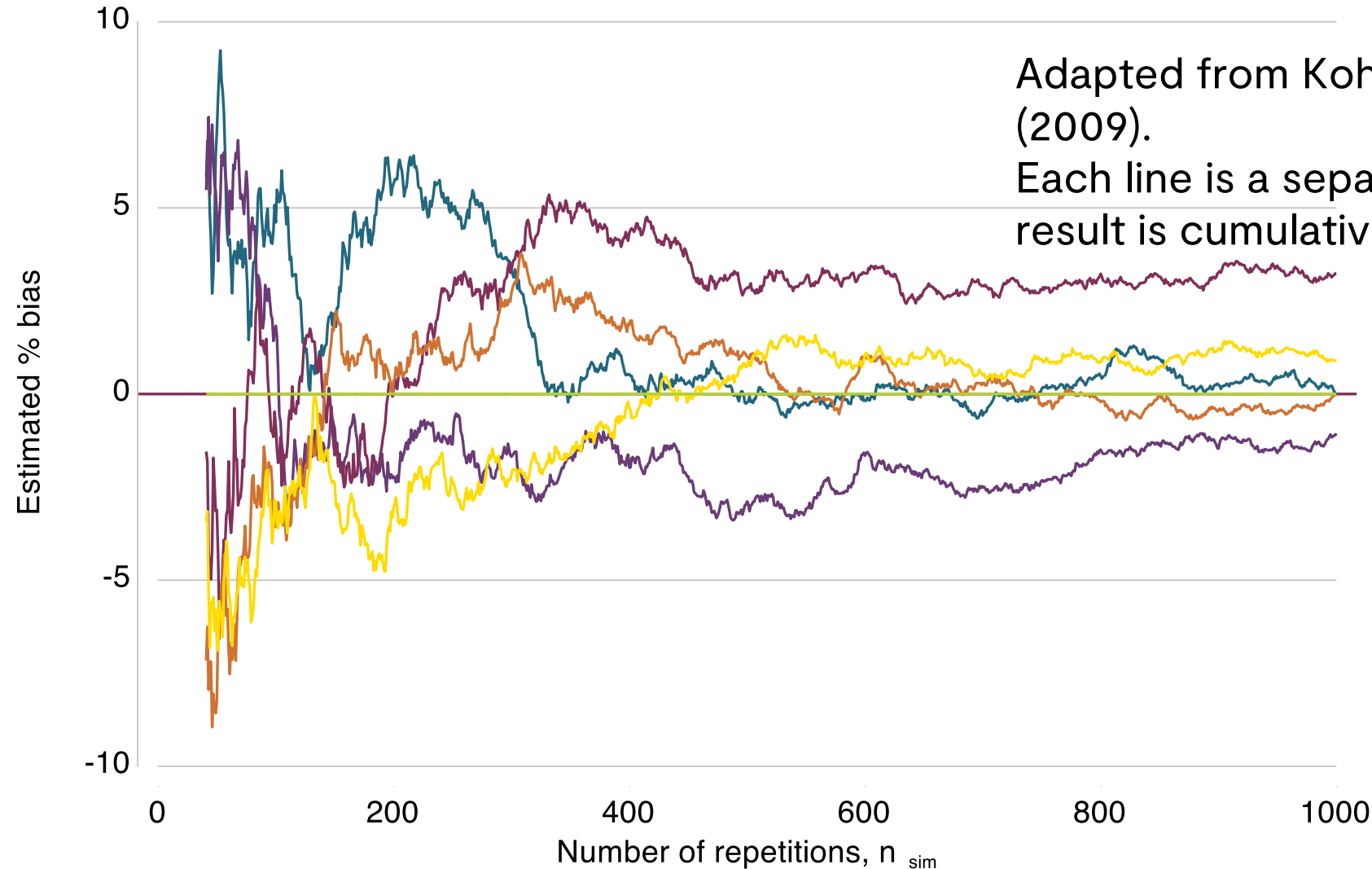
As this is an estimator, we need to work out its standard error.

- This is the Monte Carlo standard error which reflects our uncertainty about our estimate of the bias
- “Monte Carlo error”, “Monte Carlo standard error” or “MCSE”

Estimated bias is just a sample mean (minus a constant), so we use the standard formula for the standard error of a sample mean:

$$\text{MCSE}(\text{bias}) = \sqrt{\frac{1}{n_{\text{sims}}} \widehat{\text{Var}}(\hat{\theta})} = \frac{\text{EmpSE}}{\sqrt{n_{\text{sims}}}}$$

Tangent: understanding Monte Carlo error



Adapted from Kohler, Brown & Haneuse (2009).

Each line is a separate simulation study, result is cumulative estimated % bias.

Coverage: estimation and MCSE

Coverage is estimated as the proportion of confidence intervals which include the true value

- It's a proportion
- We estimate its MC error as

$$\sqrt{\frac{\text{Cover}(1 - \text{Cover})}{n_{\text{sim}}}}$$

- e.g. with $n_{\text{sim}} = 2,000$ and $\text{Cover} = 0.95$, MC error is

$$\sqrt{\frac{0.95 \times 0.05}{2000}} = 0.005 \text{ or } 0.5\%$$

How many repetitions, n_{sim} ?

The number of repetitions n_{sim} should be justified

- Basic idea: n_{sim} should be large enough to give acceptably small MC error
- e.g. in a paper by Tra My Pham she wrote “The number of simulation repetitions [$n_{sim} = 2,000$] was chosen to produce a Monte Carlo error of 0.5% on a coverage of 95%.”

Pham TM, White IR, Kahan BC, et al. A comparison of methods for analyzing a binary composite endpoint with partially observed components in randomized controlled trials. *Stat Med* 2021; 40: 6634–6650.

- Might require a Monte Carlo error of 0.1% in assessing type 1 error in a clinical trial
- Might decide bias of 0.1 on a log hazard ratio would be important, so require Monte Carlo error to be <0.03

Digression: how many repetitions?

A possible strategy:

Start with a reasonable number of reps e.g. $n_{sim} = 500$

Estimate MCSE for key performance measure(s)

- If it's small enough, stop and say so
- Otherwise, assume MCSE is proportional to $1/\sqrt{n_{sim}}$, and increase n_{sim} to achieve desired MCSE

I did this in:

Morris TP, White IR, Cro S, Bartlett JW, Carpenter JR and Pham TM. 'Comment on Oberman & Vink: Should we fix or simulate the complete data in simulation studies evaluating missing data methods?' *Biometrical Journal* [in press].

Other performance measures

I'm not going to go through other performance measures now. Look at table 6 of our Stat Med tutorial paper.

The important thing is that performance and MCSE can be estimated for many common performance measures and their MCSEs

- `simsum` (Stata, by Ian White)
- `rsimsum` (R, by Alessandro Gasparini)
- See also `interest`, a shiny app for interactive exploration of simulation results (also by Alessandro Gasparini)

Performance for our sim study

Performance measure		dgm	MI naive	(MCse)	MI intera	(MCse)
Non-missing point estimates	Scenario 1	287	.	287	.	
Non-missing point estimates	Scenario 2	291	.	291	.	
Non-missing standard errors	Scenario 1	287	.	287	.	
Non-missing standard errors	Scenario 2	291	.	291	.	
Bias in point estimate	Scenario 1	-.2243489	.0027608	-.1339763	.0050484	
Bias in point estimate	Scenario 2	.0029619	.0022275	.0161303	.0047396	
Mean of point estimate	Scenario 1	.0756511	.0027608	.1660238	.0050484	
Mean of point estimate	Scenario 2	.0029619	.0022275	.0161303	.0047396	
Empirical standard error	Scenario 1	.0467701	.0019556	.0855253	.003576	
Empirical standard error	Scenario 2	.0379978	.0015778	.0808514	.0033572	
% precision gain relative to method MI naive	Scenario 1	.	.	-70.09473	2.633839	
% precision gain relative to method MI naive	Scenario 2	.	.	-77.91277	1.722819	
...						

Performance for our sim study

Performance measure		dgm	MI naive	(MCse)	MI intera	(MCse)
...						
Mean squared error	Scenario 1	.0525122	.0012202	.0252387	.0013928	
Mean squared error	Scenario 2	.0014476	.0001379	.0067747	.0006214	
Root mean squared error	Scenario 1	.2291555	.0116183	.158867	.0275926	
Root mean squared error	Scenario 2	.0380479	.047646	.0823084	.0458593	
RMS model-based standard error	Scenario 1	.0932698	.0011362	.1247899	.0019512	
RMS model-based standard error	Scenario 2	.087537	.0010868	.1139751	.0020951	
Mean conf. interval width	Scenario 1	.3742241	.004676	.5216841	.008771	
Mean conf. interval width	Scenario 2	.3503787	.0043816	.4701889	.0085893	
Relative % error in standard error	Scenario 1	99.42167	8.684929	45.90986	6.513422	
Relative % error in standard error	Scenario 2	130.3742	9.984231	40.96867	6.401367	
% coverage of nominal 95% conf. interval	Scenario 1	27.1777	2.626017	86.75958	2.000638	
% coverage of nominal 95% conf. interval	Scenario 2	100	0	99.31271	.4843115	
% power of 5% level test	Scenario 1	1.74216	.7723012	17.07317	2.221077	
% power of 5% level test	Scenario 2	0	0	.6872852	.4843111	

What if something goes wrong?

Some tips (Brennan)

Checking a simulation study

Coding errors are **common**

- If you're lucky, the code won't run
- If you're unlucky, it will run, and you won't know for sure there's an error

Coding errors

Sometimes we can tell anyways, because results are unbelievable

- (though sometimes unlikely results are correct)

When seeing unlikely results (or sometimes even likely results!), we need to check to make sure we've not made an error

Coding errors

Based on the forthcoming paper:

“White IR, Pham TM, Quartagno M, Morris TP. How to check a simulation study. *International Journal of Epidemiology* (in press)”

Tip 1: benchmarking

Include a “benchmark” setting/method with known properties

- If results using “benchmark” are off, you know you’ve made an error

Example: in Slade and Naylor, we could include a “full data” analysis, as we **know** this should perform well

Tip 2: explore a single dataset

Look at the generated data!

Generate a single dataset

- Does it have the right proportions/means/SDs/etc?
- Are the variable ranges correct?
- Does the data structure look right?

E.g. in Slade and Naylor, do we have the right % of missing data, and the right associations between baseline covariates and the probability of missingness?

Sometimes a single **large** dataset is best (e.g. checking proportions/means/etc)

But sometimes a **small** dataset is better (e.g. to look over the data structure)

Tip 3: checking non-convergence and other errors

Sometimes analyses don't converge, or spit out ridiculous results (e.g. SE=1,000,000)

How has your program handled this? Are you summarising just the datasets for which an analysis has converged? Or are you including "SE=1,000,000"?

Can write code to "capture" non-converged/incorrect results

Tip 4: make it easy to recreate certain repetitions

Use unique identifiers (e.g. repetition number) to help you identify datasets which lead to odd results

Then, you can recreate these datasets and explore what's happened

Need to either:

- Store random number generator state at start of a repetition
- Save each simulated dataset

Tip 5: check that you haven't reset the seed

Do all your results look the same? Wonder why that is?

Maybe you've set the seed within the repetition!

This creates the same dataset every time!

Can also occur e.g. if setting a seed for multiple imputation

Tip 6: have someone else replicate your code

If you're getting weird results, ask a co-author to independently code up one of your scenarios

If results differ, compare code to see why

But, make sure they start from scratch! Any code they “borrow” from you to get started may contain errors!

Interpreting performance measures together rather than one-by-one (Tim)

- Coding a simulation study
- **Analysis of simulation studies**
- Writing up and reporting
- Reproducibility?

Performance measures are a family

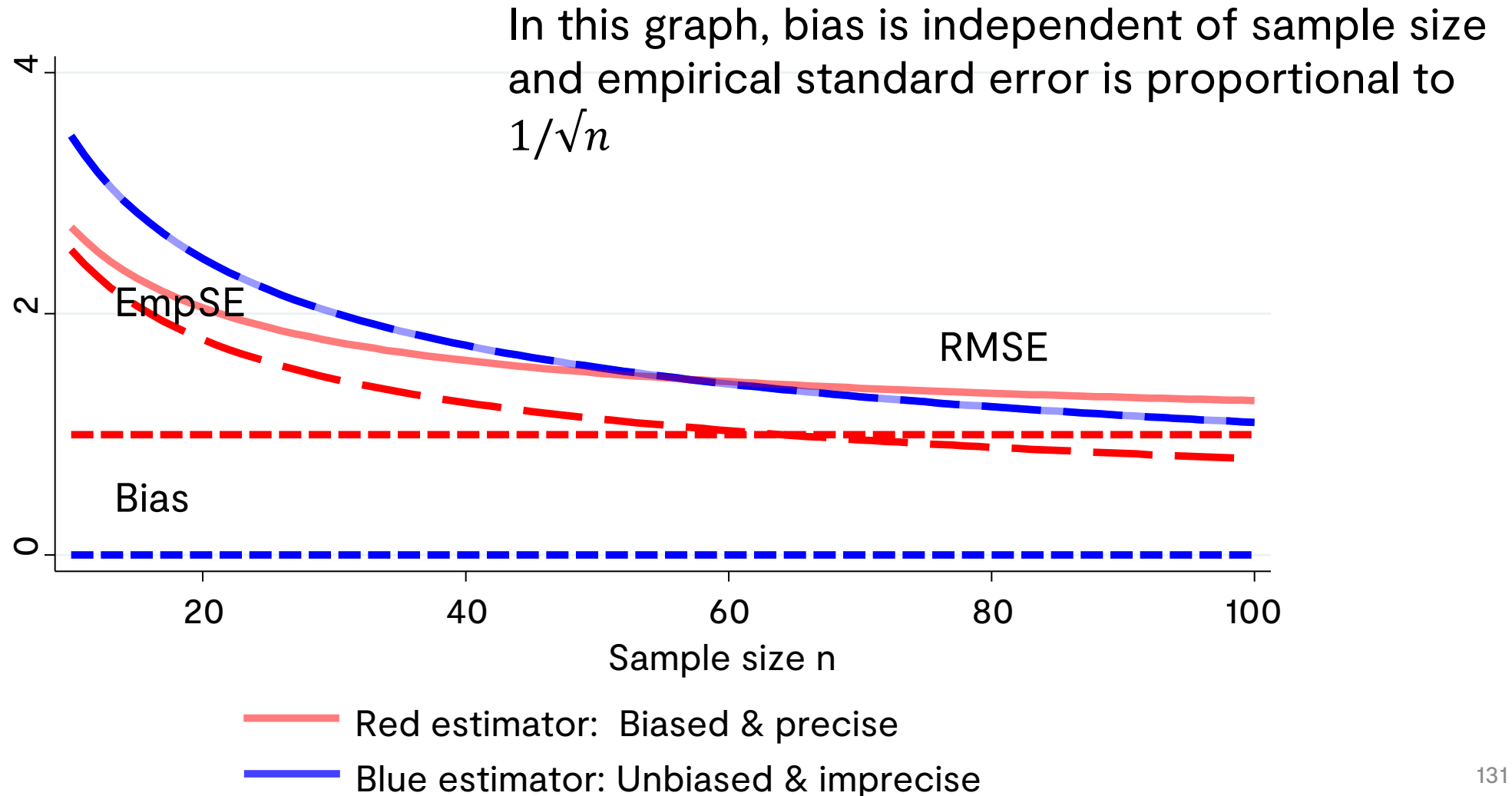
I've that if you have an unbiased estimator $\hat{\theta}$ I can have an estimator with lower variance: I just divide your $\hat{\theta}$ by a constant >1

I also mentioned that if you have bias, coverage tends to decrease

Bias away from the null tends to produce high type I error but high power

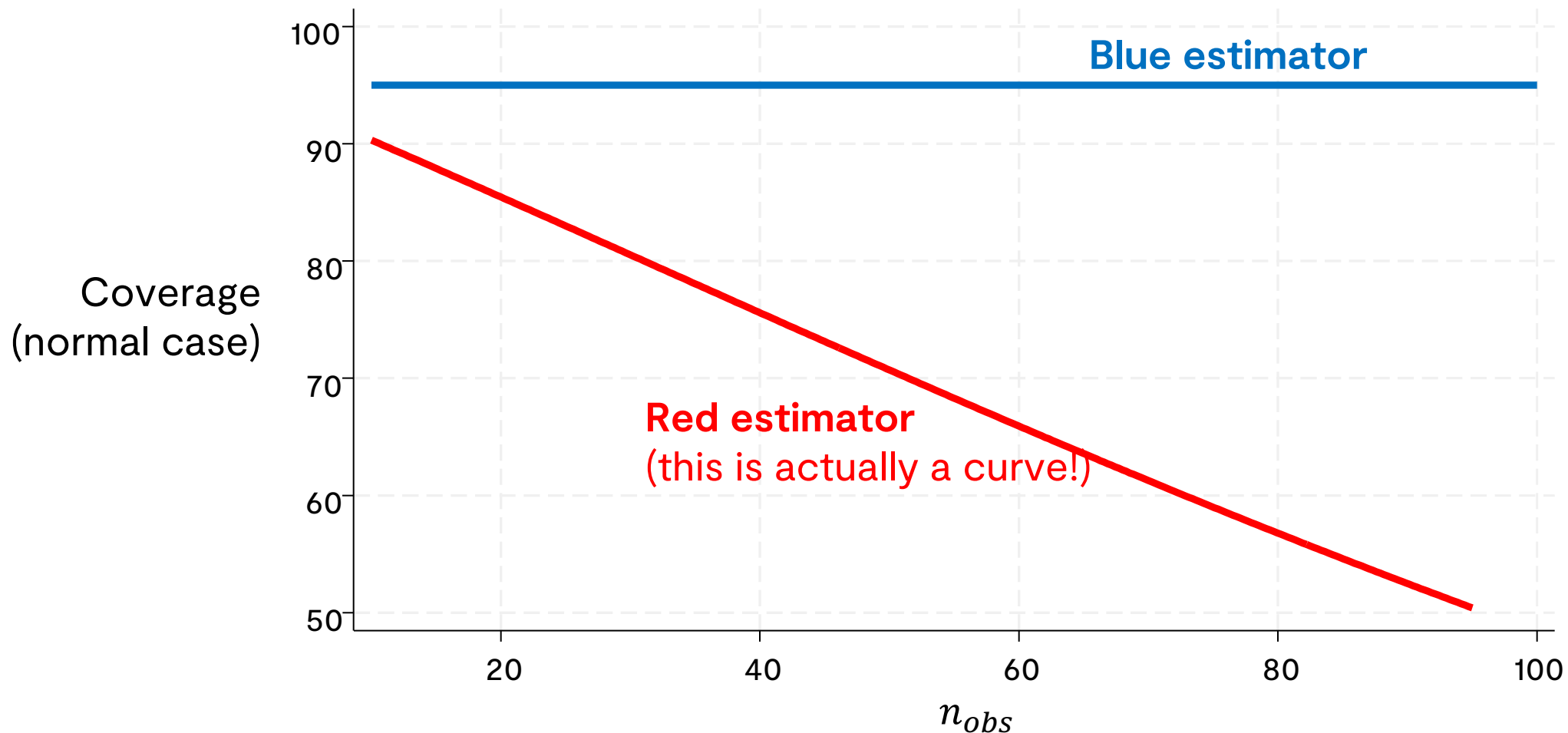
Again, we don't have time to go through how each performance measure impacts on each other one

Comparison of two estimators



Comparison of two estimators

Again, bias is independent of sample size. 95% confidence intervals are 'correct' width (when unbiased).



Tabular displays of simulation results

- Coding a simulation study
- Analysis of simulation studies
- **Writing up and reporting**
- Reproducibility?

Tabulating performance measures

There is no ‘correct’ way to do this, but here are some ideas

- Potentially four dimensions: $D \times E \times M \times P$
- Slade–Naylor had 2 DGMs, 6 estimands, 4 methods, 3 perf. meas.
- Common to have DGMs as rows, methods as columns (with more methods than DGMs, you might prefer the other way round)
- What is our comparison of interest?
 - Want to have numbers side-by-side
 - Also need to present MCSEs

There is no ‘best’ solution, but try to think this through in the planning stages

Simulation results for β_5 in a table (conditional on producing a result)

Performance measure	DGM	MI naïve	MI interactions
		Estimated performance (MCSE)	
Non-missing results	Scenario 1: $\beta_5 = 0.3$	287/300	287/300
	Scenario 2: $\beta_5 = 0$	291/300	291/300
Mean point estimate	Scenario 1: $\beta_5 = 0.3$	0.076 (0.003)	0.166 (0.005)
	Scenario 2: $\beta_5 = 0$	0.003 (0.002)	0.016 (0.014)
Mean CI width	Scenario 1: $\beta_5 = 0.3$	0.374 (0.005)	0.522 (0.009)
	Scenario 2: $\beta_5 = 0$	0.350 (0.004)	0.470 (0.009)
Coverage of 95% CI	Scenario 1: $\beta_5 = 0.3$	27.2 (2.6)	86.8 (2.0)
	Scenario 2: $\beta_5 = 0$	100 (-)	99.3 (0.5)

Estimand

Simulation results for β_5 in a table (conditional on producing a result)

Performance measure

Data generating mechs

Methods

Performance measure	DGM	MI naïve	MI interactions
		Estimated performance (MCSE)	
Non-missing results	Scenario 1: $\beta_5 = 0.3$	287/300	287/300
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	Scenario 2: $\beta_5 = 0$	100 (-)	99.3 (0.5)

Example table 1: methods for handling missing baselines (White & Thompson 2005)

Performance measure

Estimand

Data generating mechs

Table III. **Relative precision** of estimates of **treatment effect**, calculated as $1/\text{empirical variance}$, by different methods (best method = 100).

Code	Method						
		MCAR		IM1		IM2	
		$\rho = 0.3$	$\rho = 0.7$	$\rho = 0.3$	$\rho = 0.7$	$\rho = 0.3$	$\rho = 0.7$
<i>Simple methods</i>							
UN	Unadjusted	95.0	61.0	94.9	59.6	90.6	58.1
CC	Complete cases	71.3	81.4	71.7	81.9	71.8	81.9
<i>Mean imputation</i>							
I	Simple	99.9	91.3	99.8	88.1	94.9	84.7
IT	By arm	97.4	82.2	97.3	79.7	92.7	77.0
IW	Mean, weighted	99.8	99.7	99.8	97.9	95.3	95.7
<i>Missing indicator</i>							
M	Simple	99.1	90.8	99.7	91.3	99.9	91.3
MW	Weighted	99.0	99.1	99.6	99.7	99.8	99.7
<i>Bivariate outcome models</i>							
B	Simple	100	100	100	98.9	95.7	97.3
BM	Missing indicator	99.2	99.4	99.8	100	100	100

Methods

Conclude * methods are all good

+ methods good for small ρ

Example table 2: methods for multivariate meta-analysis

1 estimand

2 methods

5 performance measures

Table I. Properties of estimates $\hat{\mu}_X$ from the simulation study with $n=10$, using 1000 simulations for each run.

Run					$E(\hat{\mu}_X)$		$\text{Var}(\hat{\mu}_X)$		Z length		Z coverage		t coverage	
	I_X^2	I_Y^2	κ	ρ_i	DL	RE	DL	RE	DL	RE	DL	RE	DL	RE
1	0	0	0	0	0.002	0.001	0.007	0.007	0.370	0.355	0.961	0.960	0.987	0.977
2	0	0.3	0	0	-0.001	-0.001	0.008	0.008	0.372	0.357	0.960	0.952	0.980	0.977
3	0	0.75	0	0	0.000	0.000	0.008	0.008	0.366	0.356	0.962	0.965	0.988	0.983
4	0.3	0	0	0	-0.005	-0.004	0.012	0.011	0.418	0.408	0.936	0.928	0.966	0.960
5	0.3	0.3	0	0	0.007	0.007	0.011	0.011	0.413	0.401	0.941	0.925	0.963	0.950
6	0.3	0.75	0	0	-0.003	-0.002	0.012	0.012	0.420	0.414	0.929	0.919	0.957	0.947
7	0.75	0	0	0	0.006	0.006	0.029	0.028	0.630	0.631	0.912	0.913	0.950	0.945
8	0.75	0.3	0	0	-0.010	-0.010	0.029	0.029	0.631	0.630	0.895	0.892	0.930	0.924
9	0.75	0.75	0	0	-0.004	-0.004	0.028	0.028	0.631	0.628	0.916	0.915	0.942	0.936
10	0.3	0.3	0.7	0.7	0.006	0.006	0.011	0.011	0.406	0.408	0.927	0.925	0.958	0.959
11	0.3	0.75	0.7	0.7	-0.004	-0.004	0.012	0.012	0.403	0.402	0.919	0.925	0.953	0.957
12	0.75	0.3	0.7	0.7	0.005	0.005	0.030	0.030	0.636	0.642	0.908	0.914	0.942	0.945
13	0.75	0.75	0.7	0.7	-0.005	-0.004	0.030	0.030	0.626	0.629	0.892	0.885	0.932	0.934
14	0.3	0.3	0.95	0.95	-0.001	-0.001	0.010	0.010	0.384	0.381	0.909	0.911	0.948	0.953
15	0.3	0.75	0.95	0.95	0.002	0.002	0.011	0.010	0.392	0.392	0.938	0.949	0.963	0.970
16	0.75	0.3	0.95	0.95	0.001	0.001	0.030	0.030	0.603	0.623	0.889	0.905	0.918	0.933
17	0.75	0.75	0.95	0.95	-0.001	-0.001	0.028	0.028	0.613	0.619	0.890	0.893	0.931	0.929
18	0.3	0.3	0.7	0	0.003	0.003	0.012	0.012	0.416	0.410	0.922	0.915	0.956	0.948
19	0.3	0.75	0.7	0	0.001	0.001	0.011	0.011	0.413	0.409	0.932	0.922	0.961	0.959
20	0.75	0.3	0.7	0	0.008	0.008	0.029	0.029	0.645	0.647	0.908	0.910	0.938	0.939
21	0.75	0.75	0.7	0	-0.001	-0.001	0.028	0.028	0.616	0.622	0.903	0.906	0.938	0.935
22	0.3	0.3	0.95	0	0.003	0.003	0.011	0.011	0.415	0.410	0.930	0.927	0.963	0.956
23	0.3	0.75	0.95	0	-0.002	-0.002	0.011	0.011	0.416	0.410	0.939	0.935	0.958	0.962
24	0.75	0.3	0.95	0	0.004	0.004	0.031	0.031	0.638	0.640	0.901	0.902	0.935	0.938
25	0.75	0.75	0.95	0	0.005	0.006	0.029	0.029	0.639	0.640	0.909	0.920	0.952	0.958

25 DGMs

In each case, DL and RE denote values using the multivariate DerSimonian and Laird and REML procedures, respectively. $E(\hat{\mu}_X)$ denotes the average estimated first treatment effect and $\text{Var}(\hat{\mu}_X)$ denotes the Monte Carlo variance of these estimates. 'Z length' is the average length of a nominal 95 per cent confidence interval for the first treatment effect using the standard normal quantile and 'Z coverage' and 't coverage' denote the proportion of nominal 95 per cent confidence intervals that cover the true value, using the standard normal and t quantiles, respectively.

Alternative: summarising across DGMs?

Gutman & Rubin (2015) report summaries of coverage across many DGMs. Among other summarised performance measures, they report ‘% of DGMs for which the coverage of nominal 95% CIs was >90%’.

I dislike this approach (were you at Carolin Strobl’s talk earlier?)

We (should) want to know when a method has acceptable performance and when it has poor performance (for ‘knowns’). E.g. if I have a data set with sample size $n = 2,000$ why should I judge a method by its poor performance when $n = 200$?

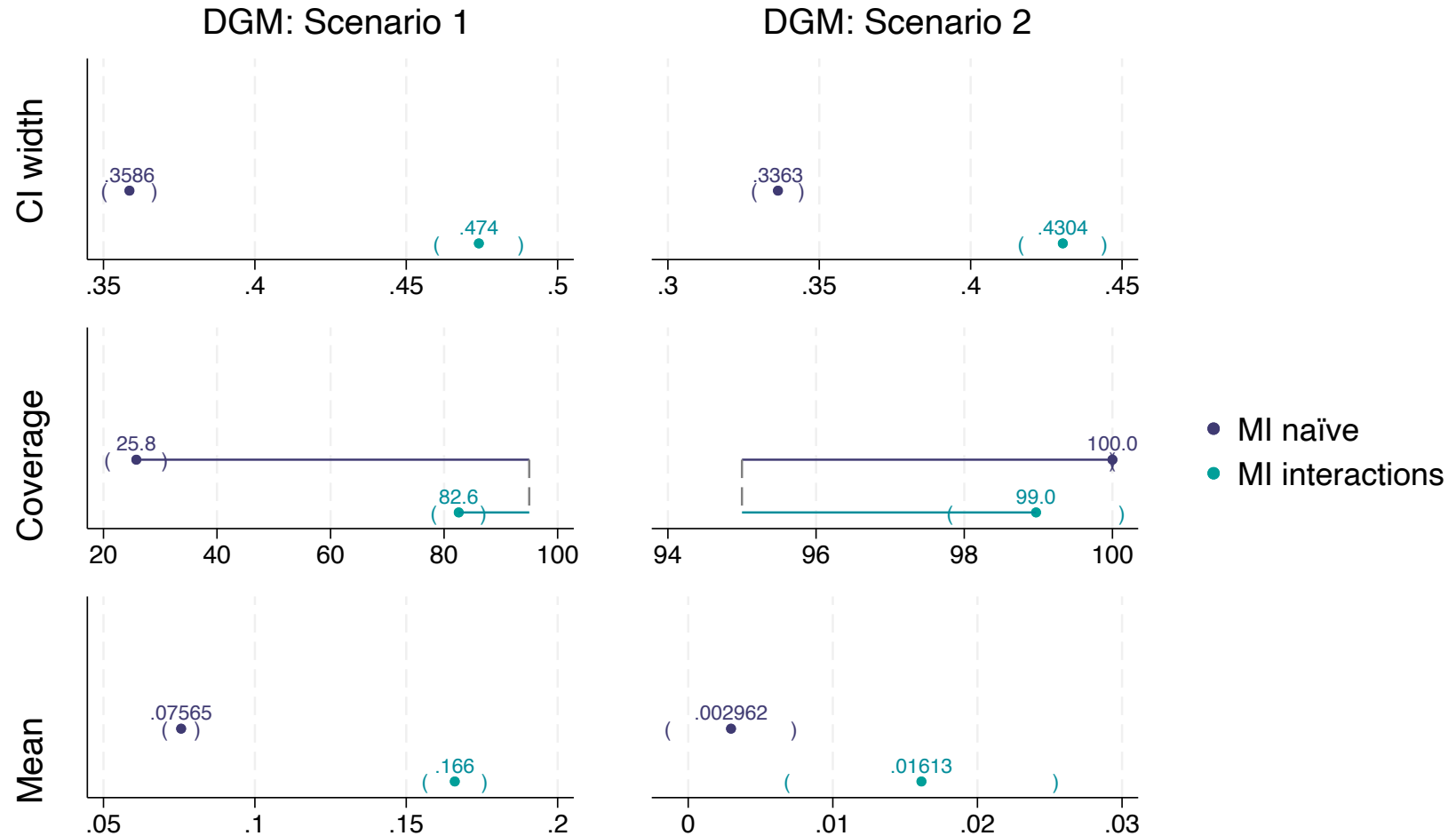
Graphical displays of simulation results

As previously, there is no ‘correct’ way to do this, but here are some ideas

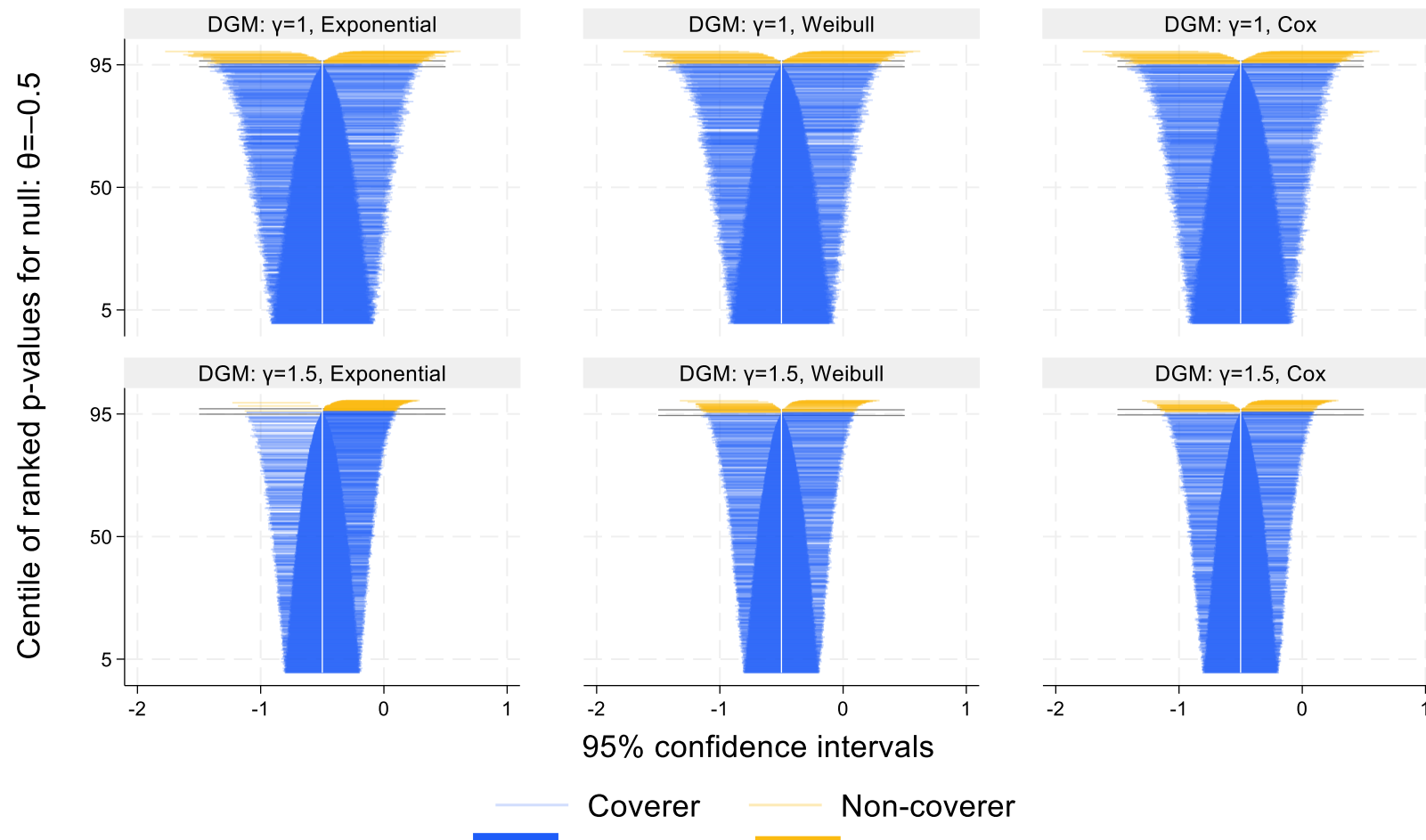
- Potentially 4 dimensions: $D \times E \times M \times P$
- Slade–Naylor had 2 DGMs, 6 estimands, 4 methods, 3 perf. meas.
- What is our comparison of interest?
 - Want to have numbers side-by-side
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There is no ‘best’ solution, but try to think this through in the planning stages

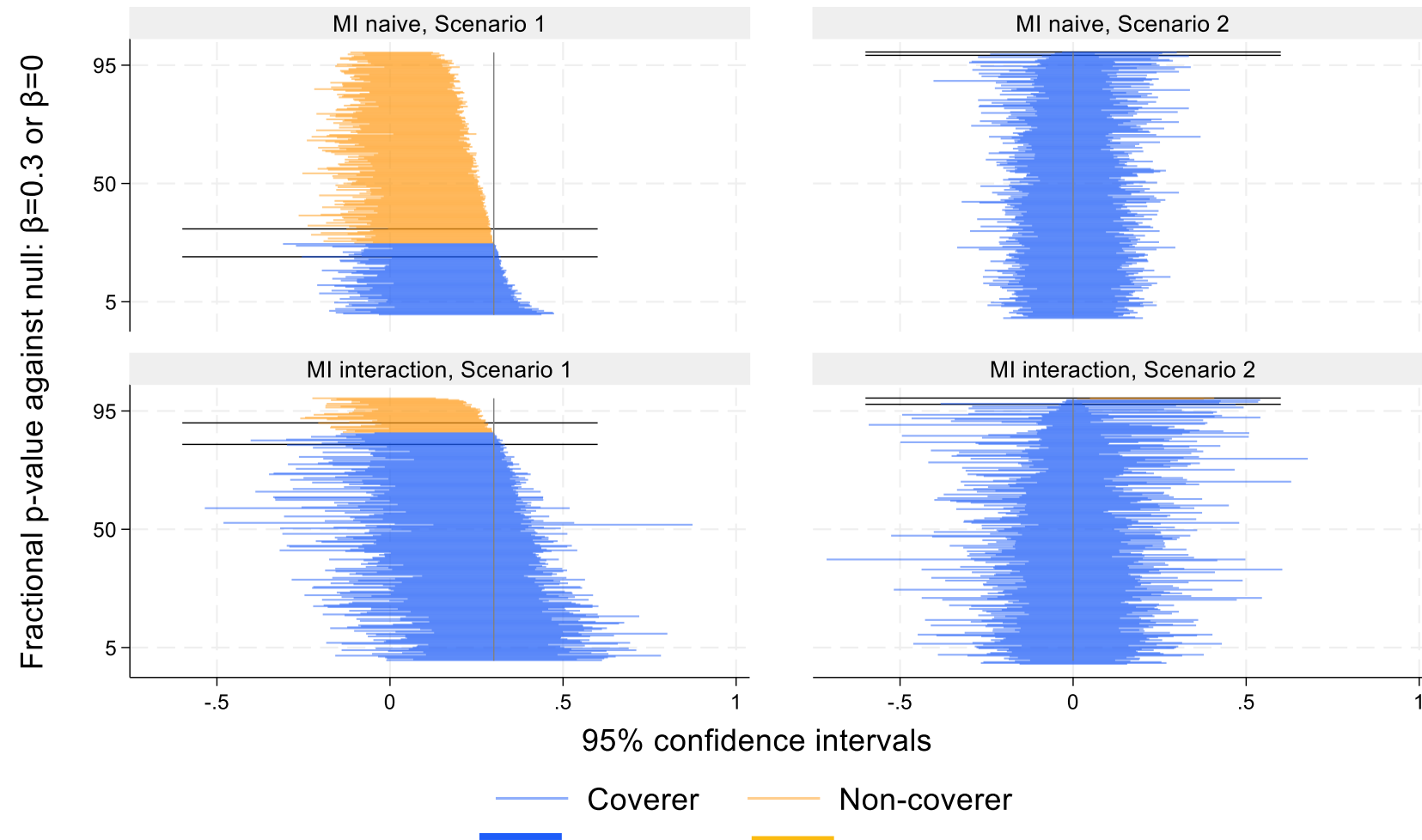
‘Lollyplot’



Zip plots to understand coverage (another sim study!)

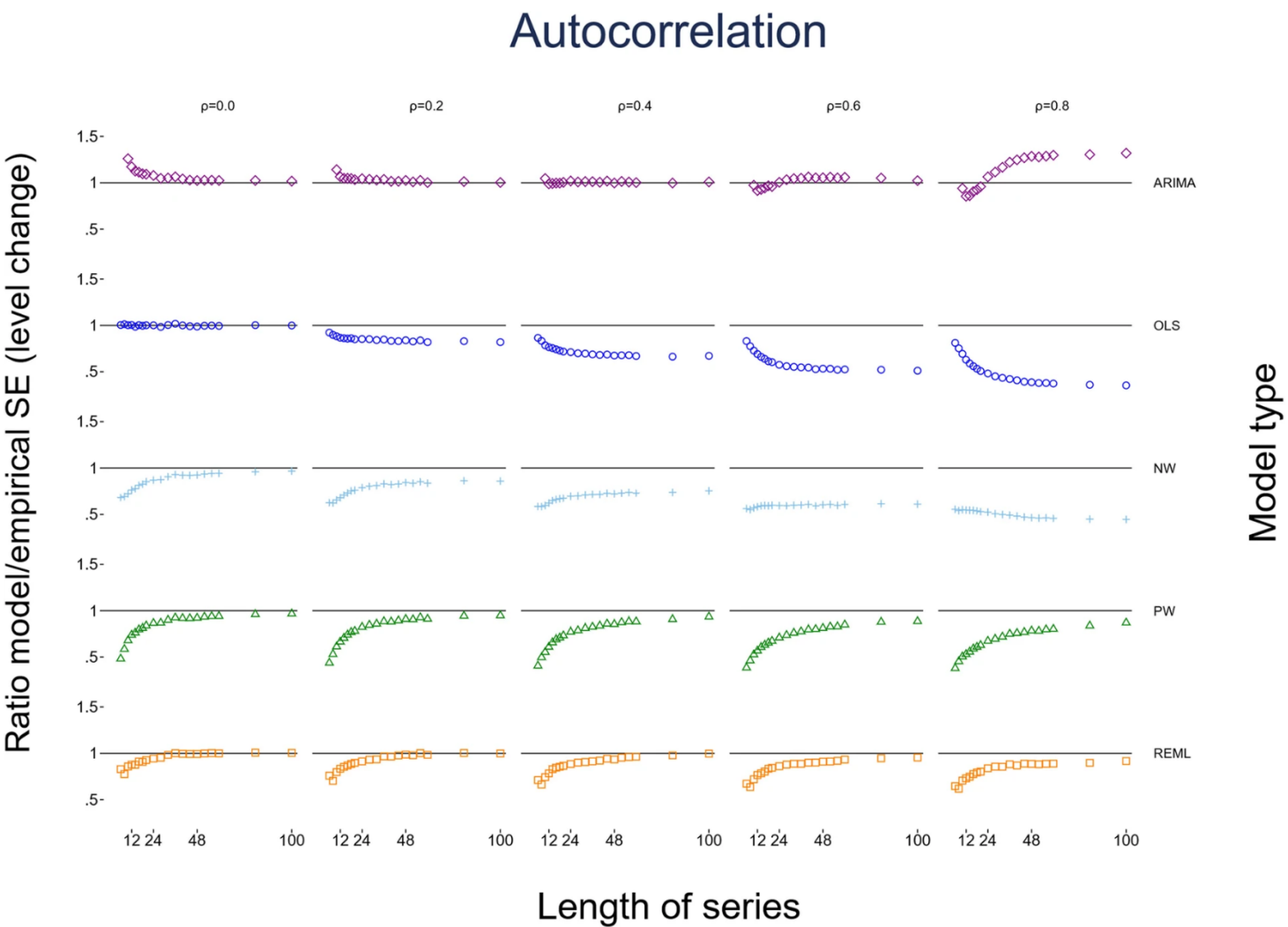


Zip plots for our simulation study



From a course participant

doi.org/10.1186/s12874-021-01364-0



Nested loop plots for 'large' factorial designs

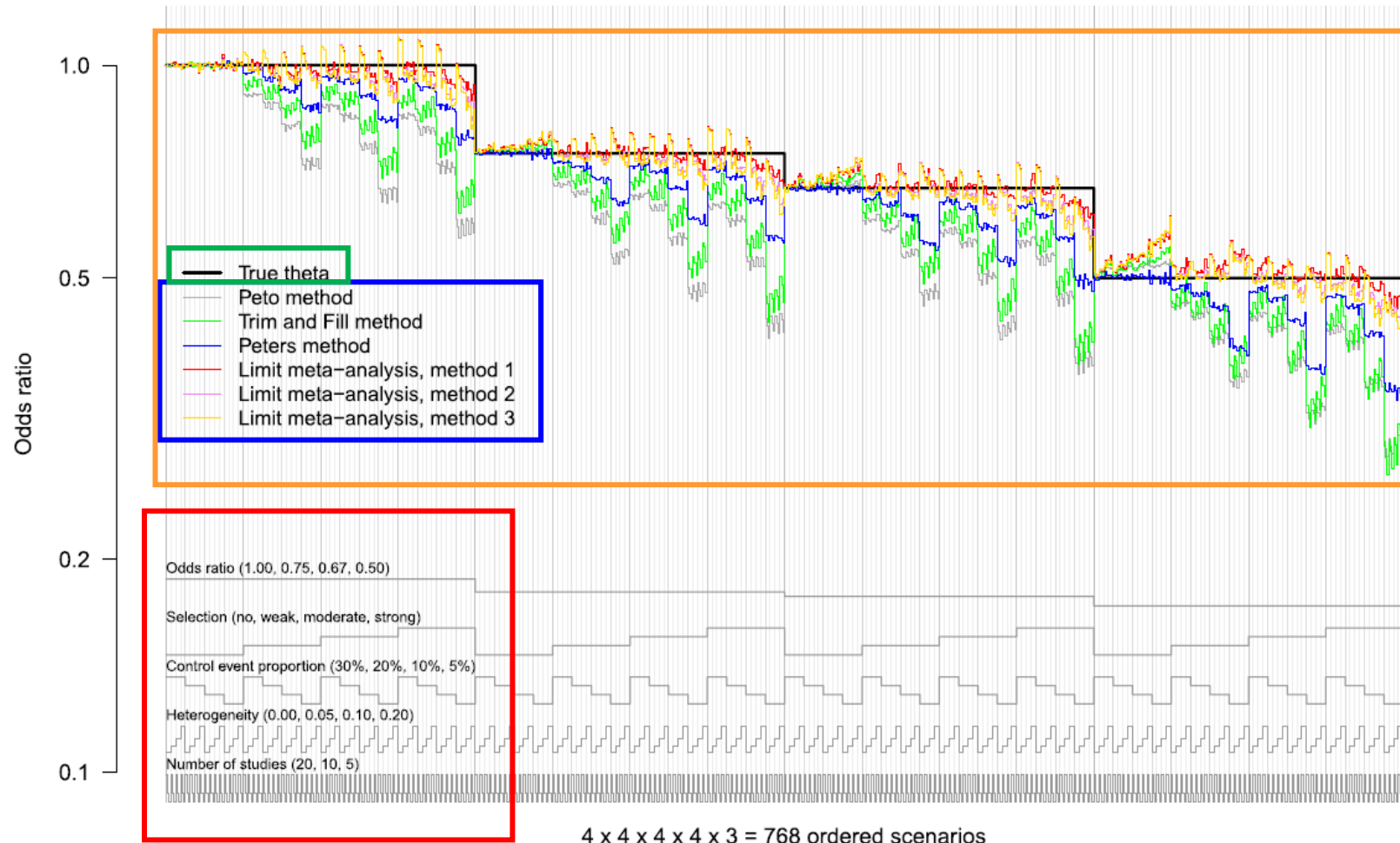


Figure 2 Nested loop plot of treatment effect estimates. 768 simulation scenarios in total, order from outer to inner loops: Treatment effect (4 levels, increasing); selection (4 levels, increasing); control event proportion (4 levels, decreasing); heterogeneity (4 levels, increasing); number of studies (3 levels, decreasing). The periodic turn of the loops is illustrated by the gray lines at the bottom of the plot.

Reporting (Brennan)

This example has been **adapted** from Slade and Naylor

- I've written this as a silly example – it is **not** taken from the paper

Questions – what (if anything) is missing that would allow you to:

- Understand what the authors **did** (enough to replicate)
- Understand what the results **mean**

Aims

‘This paper makes a fair and novel comparison of tree-based imputation models to parametric imputation models in MICE by specifying a compatible parametric imputation model as would be done in practice. We compare performance of the MICE imputation methods via simulation.’

Data generating mechanism

Part 1: Generating complete dataset

‘We generated a continuous outcome from the model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_{Int} X_2 X_3 + \varepsilon$$

X_1 , X_2 , and X_3 were generated from a $N(0, 1)$ distribution, and $\varepsilon \sim N(0, 1)$.

We set $\beta_0=0$, and $\beta_1 = \beta_2 = \beta_3 = 1$, and $\beta_{Int} = 2$ ’

Data generating mechanism

Part 2: generating missing data

‘We induced missingness via a MAR mechanism such that the total proportion of missing data is approximately 50%. In each scenario, missingness in the outcome and covariates is based on the fully observed X_3 . Subjects with $X_3 \geq 0$ have 35% missingness in other variables, and those with $X_3 < 0$ have 65% missingness in other variables, on average.’

Estimand

The estimands are: $\beta_0, \beta_1, \beta_2, \beta_3$, and β_{Int}

Methods

We applied four imputation methods: (1) predictive mean matching using default `mice` settings (PMM-Naive), (2) predictive mean matching (PMM) with an interaction term in the imputation model (PMM-Int), and (3) random forests.

The method we call “PMM-Naive” consists of the default implementation of PMM in the R package `mice`, which includes only main effects in the imputation model.

The only difference between PMM-Int and PMM-Naïve is the inclusion of an interaction term in the imputation model for PMM-Int.

We used the `randomForest` package to implement random forests in `mice`. We created 10 trees in each random forest, which is the default in the `mice` function.

Takeaways

Need to be **clear** on our aims

Explain **why** you made choices, not just **what** choices you made

What may be obvious to us isn't necessarily obvious to readers

Be precise on your methods

- Don't just say "We set 35% of data to missing" – say exactly **how** you did this!

Part B

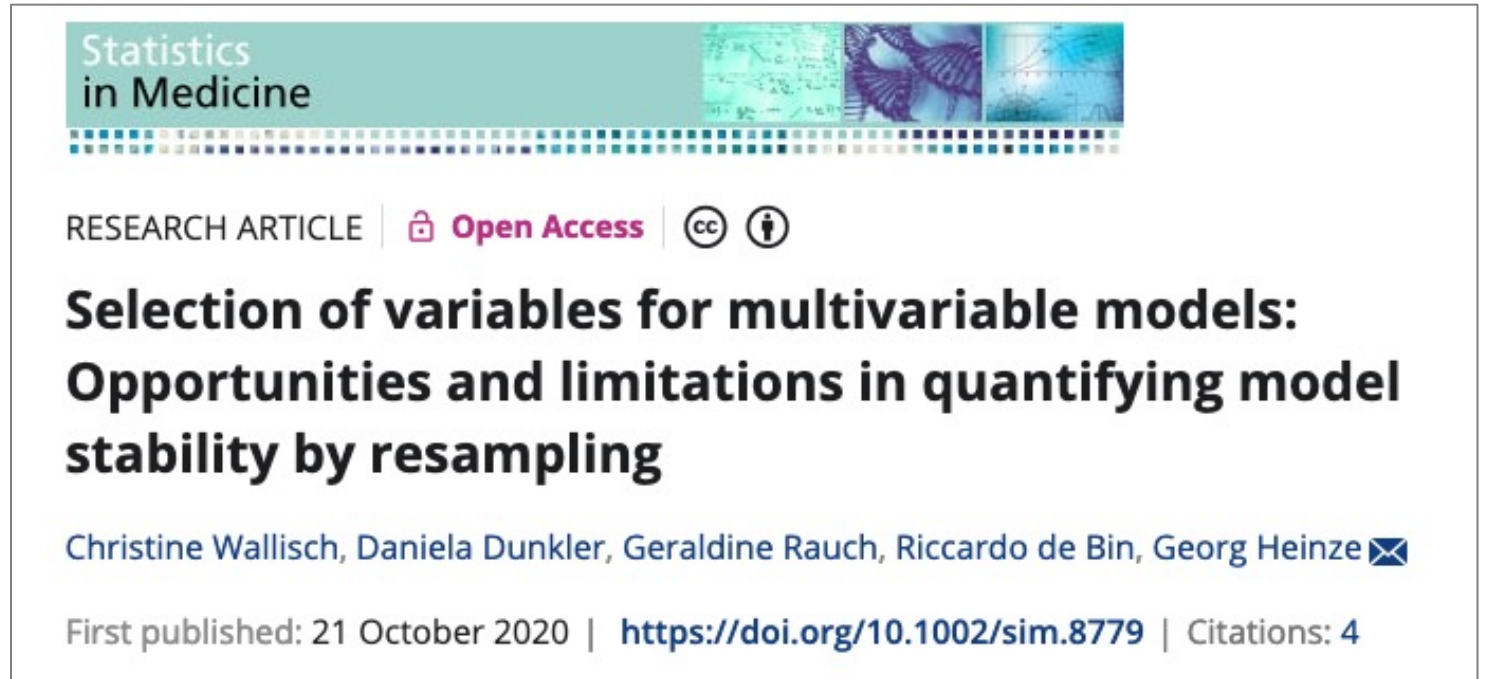
- Coding a simulation study
- Analysis of simulation studies
- Writing up and reporting
- **Reproducibility?**

Three R's?

I'm no expert on this but an interested reader. People often talk about the 'three R's': **repeatability**, **reproducibility** and **replicability**.

- We should expect to be able to **repeat** our own simulation results when we run them twice
- We should expect others to be able to **reproduce** them – but this frequently needs more information than authors give

Implementation



Ian White and I have started using ADEMPI, with I for Implementation (Wallisch, Dunkler, Rauch, de Bin & Heinze had a similar idea under ‘software’)

This is where you can put details such as choice of n_{sim} , software used, OS and environment, packages (including version and options), etc.

Replicating simulation studies



OPEN ACCESS

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It's time! Ten reasons to start replicating simulation studies

Anna Lohmann¹, Oscar L. O. Astivia², Tim P. Morris³ and
Rolf H. H. Groenwold^{1,4*}

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²College of Education, University of Washington, Seattle, WA, United States, ³MRC Clinical Trials Unit at UCL, Institute of Clinical Trials and Methodology, University College London, London, United Kingdom, ⁴Department of Biomedical Data Sciences, Leiden University Medical Center, Leiden, Netherlands

Replisims

Kim Luijken and colleagues were interested in whether seminal simulation studies actually **replicate**.

Their results are, quite frankly, alarming (<https://arxiv.org/abs/2307.02052>).

Anne-Laure Boulesteix is interested in replication where we **extend** the original simulation study beyond what the authors did to see if their conclusions hold up.

Replisims

Replicability of Simulation Studies for the Investigation of Statistical Methods: The RepliSims Project

K. Luijken^{1,2*}, A. Lohmann^{1*}, U. Alter^{3^}, J. Claramunt Gonzalez^{4^}, F.J. Clouth^{5,6^}, J.L. Fossum^{7,8^}, L. Hesen^{1^}, A.H.J. Huizing^{9^}, J. Ketelaar^{1^}, A.K. Montoya^{7^}, L. Nab^{1^}, R.C.C. Nijman^{1^}, B.B.L. Penning de Vries^{1,10^}, T.D. Tibbe^{7^}, Y.A. Wang^{11^}, R.H.H. Groenwold^{1,10^}

Pitfalls and potentials in simulation studies: Questionable research practices in comparative simulation studies allow for spurious claims of superiority of any method

Samuel Pawel  | Lucas Kook  | Kelly Reeve 

Demonstrates that it is very easy in to make a method appear better that equivalent methods.

Argue that pre-registration, data sharing and code sharing help.

Sharing code

However clear you think you have been, you will leave ambiguities in the description of your simulation study.

Sharing code helps to clear this up. People will – and do – re-run it, extend it, and adapt it. Having your code reviewed is also **an opportunity to avoid and correct errors**, as I recently discovered for a letter accepted by Biometrical Journal.

Sharing code is hard! See Darren Dahly's '*Open science is really scary y'all*': <https://statsepi.substack.com/p/open-science-is-really-scary-yall>

A final promotion

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Biometrical Journal



RESEARCH ARTICLE

On the role of benchmarking data sets and simulations in method comparison studies

Sarah Friedrich^{1,2}  | Tim Friede^{3,4} 

Statistics in Practice workshop

Some key messages

- No simulation study is perfect. There is **no objectively fair simulation** study.
- Consider the **phase of methodological research** that your simulation study intends to contribute to.
- Use **ADEMP** when planning and writing up.
- Report **Monte Carlo SEs**. If necessary, increase the number of repetitions.
- Consider your **readers**: make it easy for them to follow what you have done and where your interpretations come from
- **Share code!**

Reference list?

We weren't sure how useful this would be, but we are happy to share all the slides, where you can see snapshots of papers and pre-prints we've mentioned – and a new book on multiple imputation that Tim has shamelessly self-promoted.